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Review

Power System Decision Making in the Age of Deep Learning: A Comprehensive Review

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Abstract

Modern power systems are facing growing complexity and uncertainty due to electrification, large-scale renewable integration, and evolving consumption behaviors. These changes have pushed traditional numerical optimization methods to their practical limits in terms of scalability and real-time applicability. In response, deep learning approaches that offer fast inference and robustness to uncertainty are gaining significant attention. This paper presents, to the best of our knowledge, the first systematic review from a functional perspective of deep learning research supporting power system decision making. Taking a functional perspective, we classify neural networks into three core roles: learning to predict, learning to surrogate, and learning to optimize. In the first role, neural networks forecast exogenous uncertainties serving as the instance input to operational optimization problems. In the second role, they approximate complex physical constraints, enabling the efficient formulation of problems that would otherwise be analytically intractable. In the third role, neural networks act as learning-based optimizers that either replace or augment conventional solvers. The core purpose of this paper is to emphasize that neural networks should not simply be regarded as generic data-driven tools, but rather as models serving distinct functional roles—each with its own objectives and considerations. In this regard, we introduce diverse approaches aligned with these roles, offering conceptual foundations for principled application in practice. Such functional insights will ultimately guide the design of modular and hybrid architectures that integrate these roles, which in turn may provide the basis for developing domain-specific foundation models for power system operations.

Keywords: power system operation; optimal power flow; economic dispatch; unit commitment; probabilistic forecasting; decision-focused learning; surrogate model; learning to optimize; physics-informed

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1. Introduction

Driven by accelerating electrification across all sectors—including industry, transportation, and buildings—global electricity demand is projected to increase to as much as 2 to 2.5 times the current level by 2050. According to the International Energy Agency's (IEA) Stated Policies Scenario (STEPS), electricity demand is expected to rise from approximately 26,000 TWh in 2023 to around 50,000 TWh by 2050, which is equivalent to adding the entire electricity consumption of Japan every year [1]. At the same time, the deployment of renewable energy is being emphasized as a key strategy for achieving the Sustainable Development Goals (SDGs), and renewables are projected to account for up to 60% of electricity generation by 2030 [1,2].

This rapid surge in electricity demand, coupled with the expansion of renewable energy, presents complex challenges across all aspects of power system operation. In particular, solar and wind power exhibit high output variability depending on time and weather conditions, and their geographically distributed nature fundamentally conflicts with traditional centralized control paradigms. Moreover, the growing concentration of electricity demand in specific regions—driven by artificial intelligence (AI) data centers, electric vehicles, and heat pumps—is rapidly altering the characteristics of system loads. The intermittency and forecast errors associated with these elements undermine the efficiency and stability of power systems [3,4]. At the same time, aging infrastructure has heightened the structural risk across the power grid, where even a single contingency can escalate into large-scale blackouts [5]. In fact, from 2014 to 2023, power outages occurred on average every week in the United States, with the annual cumulative customer outage time reaching billions of user-hours [6]. These incidents extend beyond mere supply disruptions, causing significant societal impacts such as interruptions in industrial production, the paralysis of communication infrastructure, and operational failures in critical facilities like hospitals. As a result, ensuring the reliability of power system operation has emerged as a national priority—not merely a technical challenge, but a core issue linked to energy security, essential public infrastructure, and economic productivity.

These multifaceted changes in the power system environment demand a high level of flexibility and resilience [7–9]. However, integrating these capabilities into system operations entails significant cost and complexity, and conventional numerical optimization techniques have begun to reveal limitations in terms of computational efficiency and real-time applicability. Consequently, the adoption of new computational frameworks that offer both efficiency and traceability has become essential. In particular, learning-based approaches are gaining attention for their potential to effectively handle power system operation problems characterized by high uncertainty and nonlinear constraints.

Within this context, deep learning-based neural network models are being actively studied for their ability to capture complex nonlinear relationships and their adaptability to problems involving high-dimensional time-series data and embedded physical constraints. In practice, deep learning has been applied to a wide range of power system decision-making tasks, including demand and renewable generation forecasting, the approximation of dynamics, real-time optimal power flow (OPF), and unit commitment (UC) solutions. These approaches have demonstrated notable improvements in computational efficiency and generalization performance compared to traditional methods. This shift indicates a broader transition in power system operation paradigms toward learning-based decision-making frameworks and points to new strategic directions for future real-world deployment.

This paper does not limit neural networks to simple forecasting tools, but rather considers them as universal approximators capable of capturing complex computational relationships from high-dimensional inputs. It highlights the potential for strategically integrating such models across various stages of the power system decision-making pipeline. While existing studies have largely focused on individual technical components or isolated phenomena, this review takes a functional perspective, offering a comprehensive and systematic organization centered on deep learning-based approaches.

Specifically, this paper focuses on three core roles that neural networks can play in power system decision-making problems. First, neural networks can be effectively employed to forecast the uncertainty of high-dimensional time-series exogenous variables, such as electricity demand, renewable generation output, and dynamic line rating (DLR). In this role, the neural network functions as the instance of an optimization problem. Second, by approximating dynamics that are computationally expensive or analytically intractable, neural networks can significantly enhance the computational efficiency of the overall optimization pipeline. In this case, the neural network serves as part of the problem formulation. Third, neural networks can operate as learning-based optimizers that complement or replace conventional numerical solvers. Here, the neural network is used as the solver of the optimization problem.

We organize the relevant literature according to this functional classification and provide a technical comparison of the considerations required to enhance forecasting accuracy, computational efficiency, and optimization effectiveness within each approach. In doing so, it aims to offer practical insights for designing learning-based decision-making frameworks in power systems.

Based on the above perspective, this paper is structured as follows to explore deep learning-based approaches for power system decision-making. Section 2 provides an overview of analytical, non-neural network-based methods for power system operations. It begins by describing the fundamental optimization problems, including power flow analysis, OPF, and UC. It then addresses the approaches of modeling uncertainty in operational decisions, such as through stochastic optimization, and finally discusses the dynamic modeling of system behavior and stability using physics-based approaches. This section highlights the structural limitations of conventional methods, particularly in terms of scalability, real-time applicability, and robustness under uncertainty. Sections 3–5 present deep learning-based methodologies structured around three core functionalities:

- Learning to predict (Section 3) focuses on forecasting uncertain parameters (e.g., load, renewable generation, electricity prices) using neural networks. It discusses techniques to improve predictive accuracy through the consideration of data quantity, structure, and quality. It also introduces probabilistic forecasting for risk-aware decision-making, and presents decision-focused learning approaches that directly optimize downstream operational performance during the training of forecasting models.
- Learning to surrogate (Section 4) explores how complex physical constraints such as stability limits can be approximated using neural networks. This section categorizes surrogate models into regression-based and classification-based approaches, and discusses how they can be embedded into optimization formulations.
- Learning to optimize (Section 5) investigates methods in which neural networks directly approximate the solution of complex decision problems. It addresses three critical aspects: scalability to varying grid topologies via architecture design; optimality and feasibility through principled loss function design; and constraint satisfaction via post-processing and in-training mechanisms such as correction, projection layers, and implicit layers.

Then, Section 6 discusses deployment challenges that arise when these learning-based models are introduced into real-world power system operations. Finally, Section 7 concludes the paper with a discussion of integration strategies across modules, and open research challenges. Figure 1 provides a concept map and overview of this paper, highlighting the operational challenges and the three functional roles of neural networks. This serves as a roadmap for the discussions in the following sections.

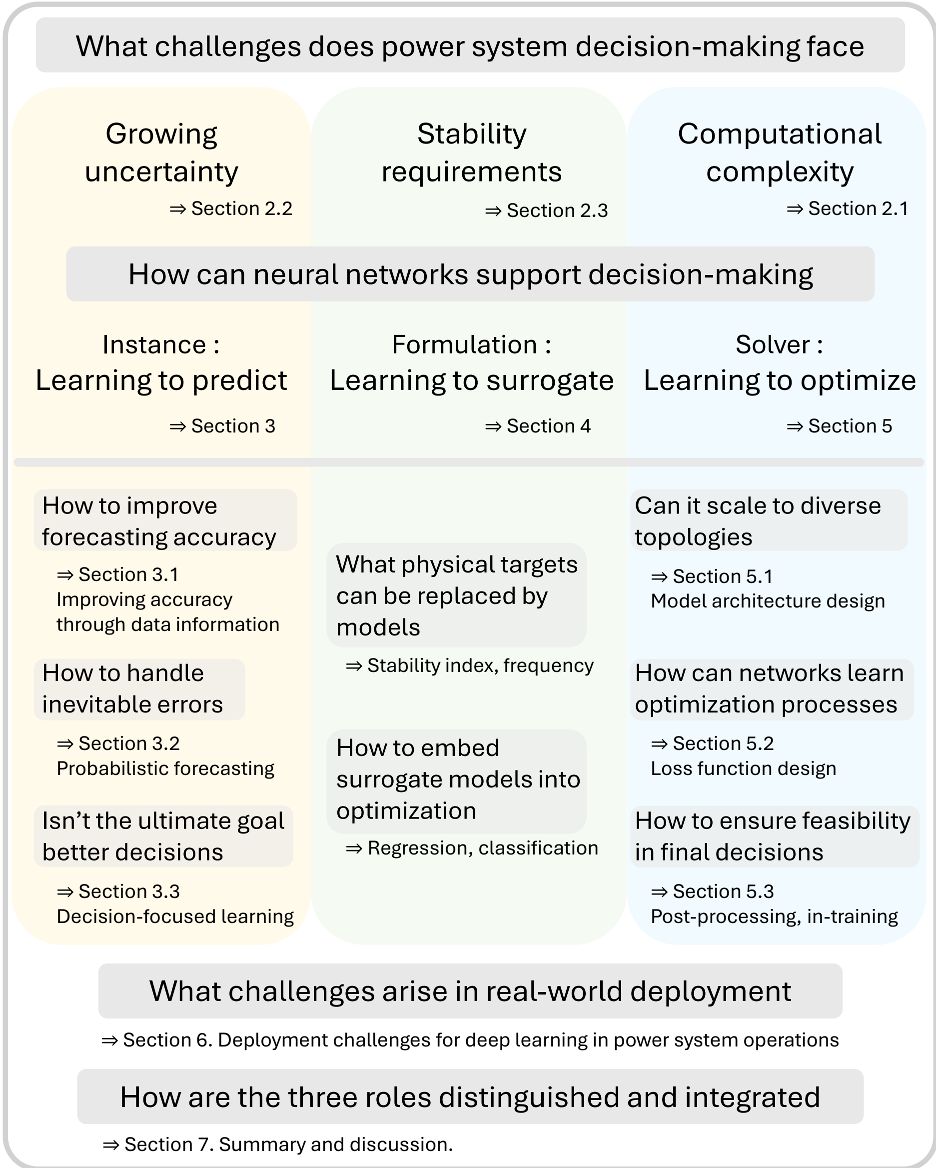

Figure 1. Concept map as an overview of this paper.

2. Operational Challenges in Power Systems and Analytical Solutions

2.1. Optimization of Power System Operation

Around 1961, as power systems began operating closer to their physical limits, concerns over transmission line overloading emerged in earnest. This marked a turning point in recognizing the need to explicitly account for the power flow problem in system operation optimization [10]. Addressing the power flow problem involves more than simply determining generator outputs—it requires identifying a feasible operating point that simultaneously satisfies various physical constraints, such as maintaining voltage profiles, adhering to transmission capacity limits, and ensuring reactive power balance. To achieve this, one needs to solve the power flow equations, a set of nonlinear equations that describe the interdependent relationships among voltage magnitudes, phase angles, and both active and reactive power flows across the entire power system.

2.1.1. Power Flow Equations: Nonlinear Constraints

These equations define the steady-state conditions under given load and generation scenarios. They consist of nonlinear algebraic relationships that link the voltage magnitudes and phase angles at each bus with the active and reactive power flows through transmission lines. In a system composed of n buses, the complex power injection S at each bus i is expressed as follows, representing the requirement that the power injection at each bus must satisfy the electrical equilibrium with the connected network:

$$S_i = P_i + jQ_i = V_i \sum_{j=1}^{n} Y_{ij}^* V_j^*$$

where V_i is the complex voltage at bus i, Y_{ij} is the (i,j)-th element of the bus admittance matrix, and * denotes complex conjugation. Depending on the type of each bus, different variables are specified, and the goal is to determine the remaining unknowns such that the electrical power balance is satisfied at each bus.

Due to the nonlinearity of power flow equations, several iterative methods have been developed. The Gauss–Seidel method is simple and memory-efficient, but it converges slowly and is highly sensitive to initial conditions, which limits its use in modern large-scale systems [11]. The Newton–Raphson method offers quadratic convergence and numerical robustness for medium- to large-scale systems [12]. However, each iteration requires the computation and inversion of the Jacobian matrix, making it computationally intensive. To reduce this burden, the fast decoupled power flow method exploits the weak coupling between active power and voltage magnitude, and between reactive power and voltage angle. However, its accuracy may deteriorate in stressed or ill-conditioned systems [13,14].

To further enhance the computational efficiency and leverage linearity, the alternating current (AC) power flow problem is often approximated by the direct current (DC) power flow equations. The DC power flow model does not account for reactive power flows, voltage magnitude variations, or the behavior of low-voltage distribution systems. Nevertheless, it remains a widely used analytical tool due to its simplicity, computational efficiency, and especially its practical effectiveness in optimization contexts. Moreover, it can provide a good approximation when voltage deviations are small and the reactance-to-resistance ratio exceeds 4 [15].

However, as discussed in the Introduction, the growing complexity of power systems and the increasing demand to consider various electrical constraints make the DC model insufficient in many cases. In such scenarios, a more accurate analysis requires the use of AC power flow models, which inevitably come with increased computational complexity.

2.1.2. Optimal Power Flow: Large-Scale Optimization

$$\text{Minimize} \quad \sum_{i \in \mathcal{G}} (c_{2,g_i} P_{g_i}^2 + c_{1,g_i} P_{g_i} + c_{0,g_i}) \tag{1}$$

$$\text{subject to} \quad P_{g_i} - P_{d_i} = \sum_{j \in \mathcal{N}} |V_i||V_j| \big(G_{ij} \cos(\theta_i - \theta_j) + B_{ij} \sin(\theta_i - \theta_j) \big), \quad \forall i \in \mathcal{N} \tag{2}$$

$$Q_{g_i} - Q_{d_i} = \sum_{j \in \mathcal{N}} |V_i||V_j| \big(G_{ij} \sin(\theta_i - \theta_j) - B_{ij} \cos(\theta_i - \theta_j) \big), \quad \forall i \in \mathcal{N} \tag{3}$$

$$P_{ij}^2 + Q_{ij}^2 \le (S_{ij}^{\max})^2, \quad \forall (i,j) \in \mathcal{L} \tag{4}$$

$$V_i^{\min} \le |V_i| \le V_i^{\max}, \quad \forall i \in \mathcal{N} \tag{5}$$

$$P_{g_i}^{\min} \le P_{g_i} \le P_{g_i}^{\max}, \quad \forall i \in \mathcal{G} \tag{6}$$

$$Q_{g_i}^{\min} \le Q_{g_i} \le Q_{g_i}^{\max}, \quad \forall i \in \mathcal{G}. \tag{7}$$

The AC OPF (ACOPF) problem aims to determine the cost-minimizing generation dispatch while satisfying the nonlinear AC power flow equations and various operational constraints, including voltage magnitudes, generator capabilities, and transmission limits [16]. Equations (3) through (9) represent the standard formulation of the ACOPF problem. The objective function seeks to minimize the total generation cost, where P_{g_i} denotes the active power output of generator i, and c_{2,g_i}, c_{1,g_i}, and c_{0,g_i} are the cost coefficients associated with generator i. Equations (4) and (5) are the active and reactive power balance equations, respectively, which must hold at every bus $i \in \mathcal{N}$. These ensure that the power supply matches demand at each node. Here, P_{d_i} and Q_{d_i} denote the active and reactive power demands, $|V_i|$ is the voltage magnitude at bus i, and G_{ij}, B_{ij} represent the conductance and susceptance of the branch between buses i and j. Equation (6) imposes the thermal limits on transmission lines by constraining the apparent power flow S_{ij}, which is defined based on both active (P_{ij}) and reactive (Q_{ij}) components. Equations (7) to (9) enforce operational limits on the system: voltage magnitudes must remain within predefined bounds at each bus, and both the active and reactive power outputs of generators must respect their physical capacity constraints.

However, due to the nonconvexity introduced by trigonometric and bilinear terms in the power flow equations, the ACOPF problem is known to be NP-hard in general [17]. This nonconvexity not only prevents guarantees of global optimality, but also poses significant challenges for conventional solvers, which may converge to local minima or fail to converge altogether, especially in large-scale or heavily loaded systems [18]. Due to the nonconvexity of ACOPF, various convex relaxations such as second-order cone programming [19], semidefinite programming [18], and quadratic programming [20] have been proposed. These methods have been shown to provide high-quality solutions and even global optima in practical cases. Their effectiveness has made convex relaxation a popular approach for addressing the challenges of ACOPF. In addition to nonconvexity, scalability poses a significant hurdle in solving large-scale ACOPF problems. As power systems grow in size and complexity—with increasing numbers of buses, generators, and transmission lines—the computational burden of solving the full nonlinear ACOPF formulation becomes increasingly prohibitive. The size of the resulting optimization problem can lead to long solution times or memory limitations, particularly in real-time or operational planning settings [21–23]. Therefore, improving the scalability of solution methods—through decomposition [24,25], parallelization [26,27], and model reduction [28–30] remains an active area of research for making ACOPF tractable in real-world systems.

2.1.3. Unit Commitment: Integer Programming

$$\text{Minimize} \quad \sum_{t\in\mathcal{T}}\sum_{g\in\mathcal{G}}(c_{2,g}P_{g,t}^2 + c_{1,g}P_{g,t} + c_{0,g}) \tag{8}$$

$$\text{subject to} \quad \sum_{g\in\mathcal{G}} P_{g,t} = \sum_{d\in\mathcal{D}} P_{d,t}, \quad \forall t \in \mathcal{T} \tag{9}$$

$$u_{g,t} \in \{0,1\}, \quad \forall g \in \mathcal{G}, \forall t \in \mathcal{T} \tag{10}$$

$$P_g^{\min} u_{g,t} \le P_{g,t} \le P_g^{\max} u_{g,t}, \quad \forall g \in \mathcal{G}, \forall t \in \mathcal{T} \tag{11}$$

$$P_{g,t} - P_{g,t-1} \le RU_g \cdot u_{g,t-1} + P_g^{\min} \cdot (u_{g,t} - u_{g,t-1}), \quad \forall g \in \mathcal{G}, \forall t \in \mathcal{T} \setminus \{1\} \tag{12}$$

$$P_{g,t-1} - P_{g,t} \le RD_g \cdot u_{g,t} + P_g^{\min} \cdot (u_{g,t-1} - u_{g,t}), \quad \forall g \in \mathcal{G}, \forall t \in \mathcal{T} \setminus \{1\} \tag{13}$$

$$z_{g,t} \ge u_{g,t} - u_{g,t-1}, \quad \forall g \in \mathcal{G}, t \in \mathcal{T} \setminus 1 \tag{14}$$

$$w_{g,t} \ge u_{g,t-1} - u_{g,t}, \quad \forall g \in \mathcal{G}, t \in \mathcal{T} \setminus 1 \tag{15}$$

$$\sum_{\tau=t-UT_g+1}^{t} z_{g,\tau} \le u_{g,t}, \quad \forall g \in \mathcal{G}, t \in UT_g \ldots T \tag{16}$$

$$\sum_{\tau=t-DT_g+1}^{t} w_{g,\tau} \le 1 - u_{g,t}, \quad \forall g \in \mathcal{G}, t \in DT_g \ldots T. \tag{17}$$

The UC problem is a fundamental optimization problem in power systems, typically formulated as a mixed-integer programming (MIP) model [31]. It determines the on/off status and generation levels of thermal units over a scheduling horizon to minimize the total generation cost. Equations (10) through (19) represent a standard formulation of the UC problem. The objective function (10) minimizes the total generation cost over the scheduling horizon, where $P_{g,t}$ is the power output of generator g at time t, and $c_{2,g}$, $c_{1,g}$, and $c_{0,g}$ are the quadratic, linear, and constant cost coefficients associated with generator g, respectively. Equation (11) enforces the power balance at each time step, ensuring that the total generation matches the total system demand. Equation (12) defines the binary commitment variable $u_{g,t}$, which indicates whether generator g is online (1) or offline (0) at time t. Equation (13) constrains the generator's output to lie within its minimum and maximum limits only when it is committed. Equations (14) and (15) impose ramp-up and ramp-down limits, respectively, to restrict how quickly a generator's output can increase or decrease between two consecutive time steps. These constraints are adjusted based on the generator's on/off status to accommodate transitions such as start-up and shut-down. Equations (16) and (17) define auxiliary variables $z_{g,t}$ and $w_{g,t}$, which represent start-up and shut-down events, respectively. These variables capture changes in the commitment status between consecutive time steps and are used in the enforcement of minimum up/down time constraints. Equations (18) and (19) enforce minimum up-time and down-time requirements. Specifically, once a generator is turned on, it must remain on for at least UT_g time steps (18), and once turned off, it must stay off for at least DT_g time steps (19). These constraints help to ensure stable and reliable generator operation while reducing unnecessary cycling.

However, the UC problem is computationally challenging due to its combinatorial nature and mixed-integer structure. The presence of binary commitment variables $u_{g,t}$ introduces a vast search space that grows exponentially with the number of generators and time steps. Additionally, the inclusion of time-coupled constraints, such as minimum up/down time limits and ramping constraints, further increases the problem complexity by linking decisions across multiple time periods [32–34]. As a result, solving large-scale UC problems to global optimality often requires significant computational resources and advanced optimization techniques. To address this computational complexity, various solution approaches have been developed. Commercial solvers such as Gurobi [35] and CPLEX [36] are widely used to solve UC problems via MIP. However, even with these powerful solvers, solving large-scale UC problems within practical time limits remains challenging. As a result, decomposition techniques, such as Lagrangian relaxation [34,37,38] and Benders decomposition [39–41], are often employed to break down the problem into smaller, more tractable subproblems. In addition, metaheuristic algorithms—including genetic algorithm [42,43], simulated annealing [44,45], and particle swarm optimization [46,47]—have been explored to obtain high-quality solutions with reduced computational effort, albeit without optimality guarantees. These approaches are particularly appealing for handling large systems or incorporating complex constraints.

2.2. Uncertainties in the Power Systems

Uncertainty in power systems arises from a variety of operational, environmental, and market-driven factors, and the most prominent uncertainty comes from the variability in load demand and intermittent renewable generation [48]. In addition, electricity market prices and equipment contingencies introduce economic and reliability risks whose stochastic behavior further compounds the complexity of planning. Each type of uncertainty presents distinct statistical and temporal characteristics that challenge conventional planning and operational strategies.

The uncertainty of load demand stems from the difficulty in predicting consumer behavior, economic activity, and weather influences. The load exhibits strong periodic patterns, but short-term deviations such as unexpected behavioral changes or meteorological fluctuations can result in significant forecast errors [49]. The uncertainty of renewable generation, particularly from wind and solar sources, presents a similar major challenge. Renewable output is intrinsically dependent on meteorological conditions such as wind speed and solar irradiance, which cannot be forecast with perfect precision [50]. Wind energy exhibits rapid and sometimes large fluctuations on multiple time scales, and solar generation is subject to sudden changes caused by cloud cover [51].

2.2.1. Modeling Uncertainty

To deal with these uncertainties, early work focused on providing accurate point forecasts [52,53]. However, because future conditions are inherently uncertain, perfect load and renewable forecasts may not be possible. Researchers therefore moved beyond point forecasts and focused on modeling uncertainty statistically. Many studies assumed that forecast errors or fluctuations follow simple parametric distributions to facilitate analytical tractability [54]. In renewable generation, domain-specific distributions have been used: wind speeds are commonly modeled by a Weibull distribution and solar irradiance by a Beta distribution, providing the probabilistic characterizations of their variability [55]. Time-series techniques such as the autoregressive integrated moving average and its variants have also been widely applied to capture temporal patterns in load demand [56]. These statistical methods are transparent and require limited data, but often assume linearity, normality, or stationarity assumptions that can be unrealistic under the high variability of modern power system uncertainties. Such limitations have motivated the development of more advanced data-driven approaches, e.g., deep learning models, that better capture the true underlying distributions.

2.2.2. Decision Making with Uncertainties

Decision making in power systems should explicitly account for uncertainties in both operational timescales (e.g., UC, OPF) and long-term planning (e.g., transmission expansion, generation capacity investment). A variety of mathematical optimization approaches have been developed to handle these uncertainties. Prominent methods include scenario-based optimization, robust optimization, and chance-constrained optimization. Each offers a different balance between optimality and risk, and all have seen extensive use in the power system operations and planning literature.

Scenario-based optimization is a stochastic programming approach that represents uncertainty by a set of discrete scenarios and optimizes decisions between these scenarios (often minimizing expected cost or risk). In power systems, this method is widely used for operational planning problems like stochastic UC and security-constrained (SC)-OPF, where multiple realizations of renewable output or demand are considered simultaneously. For example, a scenario-based stochastic UC co-optimizes generation schedules and reserves under various wind and load scenarios and has been shown to outperform deterministic reserve rules [57]. In long-term planning, scenario-based models help evaluate capacity expansion and transmission planning under different future conditions (e.g., policy or demand growth scenarios) [58]. This approach explicitly models uncertainty and can reduce expected operating cost and enhance system reliability while providing insights into the economic impacts of variability. However, the precision of the results depends on the quality and number of scenarios; a large set of scenarios better captures uncertainty but drastically increases the computational requirements [59]. Therefore, selecting and weighting representative scenarios is critical and scenario reduction techniques are often needed to balance the accuracy and tractability of the solution [60].

Robust optimization is a worst-case approach that seeks feasible solutions for all values of the uncertain parameters within a prescribed uncertainty set. Rather than relying on probability distributions, the decision maker defines bounds or sets (e.g., ranges of wind output or load), and the optimization then minimizes cost under the worst-case realization in that set [61]. In power systems, robust methods have been employed in operations such as UC under wind uncertainty and ACOPF [62,63]. Robust optimization provides strong reliability guarantees. However, its solutions can be overly conservative and costly, as preparing for the absolute worst-case scenario may result in the dispatch of excessive reserves or the over-investment in capacity.

Chance-constrained optimization imposes probabilistic constraints, requiring that system constraints (e.g., generation limits and line flows) are satisfied with a high predefined probability instead of under every possible outcome. In power systems, chance-constraints have been applied to operational problems such as OPF and UC [64,65]. This approach provides a principled way to balance cost and reliability by allowing a small, quantifiable risk of constraint violation. The chance-constrained formulation shows close alignment with industry practices such as probabilistic reserve dimensioning in the European Network of Transmission System Operators for Electricity (ENTSO-E) [64]. However, solving chance-constrained problems can be challenging because the probabilistic constraints are generally non-convex and difficult to evaluate directly. Thus, practical solution methods are based on approximations.

2.3. Power System Dynamics and Stability

Dynamic modeling plays a central role in power system analysis and control, as it captures how the system responds to disturbances, control actions, and changing operating conditions [66]. These models are essential for assessing system stability, designing frequency and voltage controllers, and simulating future operating scenarios [67,68]. To support such tasks, conventional engineering practice has relied on physics-based formulations grounded in well-established electrical and control theory. The dynamic behavior of key components—including synchronous machines, power electronic converters, control loops, and filter networks—is typically described by differential-algebraic equations (DAEs) derived from physical laws and system specifications [66,69]. These models form the foundation for small-signal and transient stability analysis, controller tuning, and time-domain simulation, and continue to serve as standard tools in planning and operational studies. As the grid evolves to incorporate more inverter-based resources and increasingly complex dynamics, the need for accurate, adaptable, and transparent dynamic models is greater than ever [70].

Despite their importance, both accurate dynamic modeling and reliable stability assessment remain fundamentally challenging in practice. Key parameters—such as inertia constants, damping ratios, and control gains—are frequently undisclosed or outdated, and often fail to reflect real-world behavior due to aging, environmental factors, or undocumented modifications [71–73]. For synchronous generators, offline testing methods are commonly employed to estimate key dynamic parameters when manufacturer data are unavailable. These traditional methods include short-circuit tests [74], standstill frequency response [75], open circuit frequency response [76], and least squares [77], which are typically conducted when the generator is out of service. However, these procedures are often time-consuming, require specialized equipment, and cannot be performed during operation, limiting their practicality for real-time or online applications [76].

On the other hand, even with well-estimated parameters, real-time stability assessment remains computationally intensive. Conventional methods often require solving large-scale dynamic equations or performing eigenvalue analysis at a high temporal resolution, which poses a significant burden for online implementation [78,79]. To alleviate the computational burden, a wide range of approaches have been proposed. For instance, the authors of [80] developed a real-time voltage stability assessment method based on Thévenin equivalent impedance using phasor measurement unit (PMU) data, enabling efficient monitoring without full system modeling. Similarly, the work in [81] demonstrated the feasibility of real-time transient stability simulation using a partitioned approach on parallel digital signal processors, highlighting the advantage of distributed computation for solving large-scale dynamic equations efficiently. To further reduce computational cost, the study in [82] introduces a hybrid direct time-domain approach for transient stability assessment, simplifying multi-machine systems into a single machine equivalent and enabling the fast estimation of stability margins and critical contingencies with only a few short simulations. However, these methods remain effective only when system dynamics are explicitly modeled with well-defined parameters, which limits their robustness under model uncertainty or varying operating conditions [78,81].

2.4. Emerging Neural Architectures

Traditional analytical approaches are increasingly limited in handling the scale and complexity of modern power systems. Beyond the challenges of large-scale dimensionality, nonlinear formulations, and integer programming, the growing diversity of uncertainty sources and the strict requirements for stability make conventional methods inadequate. These limitations have driven interest in advanced neural network paradigms, and more recently, discussions around the potential of foundation models tailored to power systems have emerged [83]. A foundation model refers to a neural framework pre-trained on large, diverse datasets and then fine-tuned for a wide range of downstream tasks. Such models hold significant promise for addressing complexity and uncertainty by allowing stakeholders to adapt pre-trained models to specific operational needs or proprietary data under strict security requirements. Although still at a conceptual and exploratory stage for power system applications, this adaptability also provides robustness in data-scarce environments, which are common in power grids.

For complex engineering systems like power systems, embedding domain knowledge into neural architectures is a natural evolution beyond purely data-driven black-box methods. Recent advances in fields like fluid dynamics, materials science, and control theory highlight the potential of physics-informed neural operators [84]. Lu et al. [85] linked the universal approximation theorem for operators with deep neural networks, showing that neural operators can learn a wide class of explicit and implicit operators across diverse applications.

In this review, we describe these concepts in general modeling terms—such as loss functions, model outputs, and network architectures—rather than focusing on emerging specific model names. In essence, a physics-informed neural operator treats the result of a physical operation as the model output and designs an appropriate loss function such that the approximator embodies the meaning of the underlying operator. Specifically, the physics-informed neural network named PINN [86] can be interpreted as an approximator of DAE solutions. By treating neural network outputs as both function values and their derivatives with respect to inputs, the loss function penalizes deviations from the governing equations, thereby guiding the network to operate as a DAE solver for the underlying physics.

In power systems, physics-informed graph convolutional networks have been proposed in [87], where grid topology is embedded by representing buses as nodes and transmission lines as edges—naturally aligning the architecture with the physical system structure. Notably, the notion of being "physics-informed" is not fundamentally unique; it simply reflects the fact that the relevant domain happens to be physics. Neural networks that embed domain-specific information have already been developed in many other areas. High-performing models—such as transformers, diffusion models, and recurrent neural networks—inherently incorporate inductive biases and structural priors that reflect domain knowledge. This broader perspective provides an intuitive way to understand the evolution of advanced models across domains. Chen et al. [88] exemplify this by integrating multi-factor attention mechanisms with physically interpretable queries and graph topology representations into transformer architectures, enabling the robust characterization of sectional power system states. By adopting this broader perspective on advanced models, neural networks can be reinterpreted as approximators endowed with physical meaning through loss functions, inductive architectures, and interpretable outputs—thereby supporting their systematic application in power system operations.

3. Learning to Predict: From Uncertainty Inputs to Operational Insight

Deep learning-based models have emerged as core technologies in a variety of forecasting tasks due to their ability to model nonlinear relationships [89,90]. Rather than explicitly modeling the various factors influencing the target variables through physical or statistical methods, these models address forecasting problems by approximating unknown, complex nonlinear relationships in a data-driven manner [91]. This approach allows for the more flexible incorporation of high-dimensional features, enabling high prediction accuracy even in situations where explicit modeling is challenging. These advantages are particularly relevant in the power systems domain, where variables such as demand, renewable generation, and electricity prices are high-dimensional and highly uncertain. As a result, deep learning-based forecasting models have been actively studied in this field [92–95].

However, from the perspective of power system operation, the output of the predictive model is not a simple outcome, but a major parameter directly input to the operational optimization problem at a later stage. Once a forecast is made, it becomes the basis for key decisions such as generator dispatch, charge–discharge scheduling, and market bidding, all of which are tightly linked to actual system operation. The challenge, however, lies in the fact that forecasting errors are inevitable, and their impact on downstream optimization outcomes is not straightforward to quantify [96–98]. To ensure reliable operation in the presence of such forecast errors, the traditional approach is to use stochastic optimization, as discussed in Section 2.2. In this context, neural network-based forecasting models need to go beyond point predictions and quantify uncertainty in the output, typically by generating a set of scenarios or a probability distribution over the forecast.

Meanwhile, an integrated approach that considers even the quality of decision making when learning predictive models is also emerging. In other words, rather than treating prediction as a separate step from optimization, it is a method of designing or learning a predictive model based on the performance of the final decision problem. Accordingly, this section reviews (1) various approaches aimed at improving the forecasting accuracy itself, (2) probabilistic forecasting techniques for enabling robust operational optimization, and (3) decision-focused learning strategies that directly optimize for downstream decision quality.

3.1. Improving Forecast Accuracy Through Data- and Structure-Aware Neural Modeling

Research on predicting uncertainty factors in power systems started with a structure that uses multi-layer perceptron (MLP) for prediction, and demonstrated the predictability through algorithm research for learning stabilization [99–102]. However, advancing forecasting models goes beyond simply designing more complex architectures [103]. Increasing model complexity alone not only raises the computational burden of training but also heightens the risk of overfitting [104]. Therefore, a multifaceted understanding of the data and the model to be used should precede model design [105]. First, the data should be considered in terms of quantity to construct a learning algorithm effective for generalization. The form of structural connectivity in the data is connected to the structure of the model, and understanding the quality of the data enables the design of appropriate pre-processing steps. In this section, we systematically summarize various approaches attempting to improve prediction accuracy based on this understanding of the data.

3.1.1. The Quantity of the Data

A study that conducted a quantitative analysis of the relationship between data complexity and model capacity [106] theoretically demonstrated that the more domination data there is in the data, the more model capacity is required. Additionally, estimating the actual model complexity is a key challenge, as model capacity is associated with generalization performance [107,108]. These studies demonstrate the need to adjust the amount of data and consider efficient model capacity through hyperparameter tuning algorithms.

Lai et al. [109] employed a deep neural network (DNN)-based regression model for monthly load forecasting. To enhance the model's generalization performance and improve prediction accuracy, they increased the number of training samples using historical data augmentation. Ryu et al. [110] applied DNN to individual load forecasting, moving beyond aggregated electricity consumption forecasting in the context of demand-side energy management. Noting that the potential for overfitting depends on data volume, they trained the model on three years of load data and achieved high accuracy without overfitting. Malakouti et al. [111] utilized K-fold cross-validation and grid search to identify appropriate hyperparameters for predicting solar farm power generation. Zhou et al. [112] proposed sequence model-based optimization, performing hyperparameter tuning tailored to time-series models for electricity price forecasting.

On the other hand, meta-learning approaches are attracting attention as a way of maintaining the generalization performance of the model in the face of a lack of data. Meta-learning actively utilizes previous learning experiences as a way of "learning to learn" [113] to quickly learn new tasks. Specifically, Finn et al. [114] proposed a model-agnostic meta-learning method that learns "good initialization" that allows for fast learning with only a small amount of data and a small number of gradient descent, given a new task.

Meta-learning approaches have proven effective for household-level energy prediction even in limited data environments, and can quickly adapt to new environments with minimal data and significantly improve short-term load prediction accuracy [115–117]. Beyond household-level load prediction, meta-learning techniques have also been successfully applied to renewable energy prediction tasks such as solar and wind power generation prediction. Ren et al. [118] proposed a model consisting of a transferable learner that learns the knowledge of the source solar power plant through transfer learning and an adaptive learner that performs meta-learning-based fast adaptation. The proposed model is based on the two pre-trained learners, and effective fine tuning is possible with only a small amount of data from the target power plant, thereby achieving high accuracy solar power prediction performance. Chen et al. [119] proposed WG-Reptile, an improved version of the original reptile algorithm [120] to effectively leverage knowledge from source wind farms, resulting in more accurate predictions at the target site. The method incorporates within-task sample allocation based on contextual scenario techniques to allocate training samples, and gradient conflict alleviation using cosine similarity to mitigate knowledge conflicts between source tasks, thereby improving meta-learning-based forecasting performance.

The aforementioned studies have strengthened adaptability to limited or unseen datasets by selecting appropriate training algorithms—such as early stopping, hyperparameter tuning, and meta-learning—based on the amount of available data.

3.1.2. The Structural Dependencies in the Data

When the data itself exhibits spatio-temporal connectivity, it is crucial to adopt model architectures that incorporate inductive biases aligned with these characteristics. For instance, when dealing with time-series data, recurrent structures such as recurrent neural network (RNN), gated recurrent unit (GRU), and long short-term memory (LSTM) are particularly effective [121–123]. In cases where the data contains spatial dependencies, models that can capture local or topological relationships—such as convolutional neural network (CNN) or graph neural network (GNN)—tend to perform better [124,125].

To capture the temporal characteristics of uncertainty in power systems, various RNN architectures have been applied, including LSTM networks—which incorporate three gating mechanisms to learn long-term dependencies—and GRU, which reduce computational complexity [126–130]. Among these, Shi et al. [131] proposed a pooling-based deep RNN for residential load forecasting. This model incorporates spatial information among neighboring households via a pooling layer to compensate for the limited temporal context captured by standard RNN, thereby enabling more personalized predictions for household-level loads. Jeong and Kim [132] introduced the space–time CNN for short-term solar power forecasting across multiple photovoltaic (PV) sites. In this approach, time-series data from each site are arranged into a space–time matrix based on their geographical locations. A greedy adjoining algorithm, based on pairwise distances between PV sites, was devised to determine spatial adjacency, enabling the CNN to effectively learn spatio-temporal correlations. The work of [133] proposed a GNN-based forecasting framework for large-scale datasets involving thousands of PV sites. By combining attention mechanisms with spatio-temporal learning, the model is designed to be robust to spatial and temporal data gaps. Importantly, the message-passing operations inherent in the GNN architecture allow for scalable learning across a large number of nodes, while a parallel LSTM component enables the integrated learning of both temporal and spatial dependencies.

Beyond spatio-temporal connectivity, recent research has actively explored approaches that reflect the intrinsic frequency components of time-series data. Time-series signals generally consist of multiple frequency components, and decomposing these components and modeling their unique characteristics can contribute to improved forecasting performance. Wang et al. [134] applied wavelet transform to decompose PV power generation time-series into multiple frequency components, enabling a CNN to extract nonlinear and invariant features from each component. Choi et al. [135] proposed a two-stage neural network in which residual network was used to target the trend of data (slow-changing), while LSTM was employed for the seasonality of data (rapidly fluctuating). This approach effectively predicted load data exhibiting both regularity and irregularity. Oreshkin et al. [136] introduced N-BEATS, a forecasting architecture that achieved state-of-the-art predictive capability without relying on RNN. By leveraging the statistical characteristics of trend and seasonality, they designed an interpretable forecasting model based on residual MLP blocks. Su et al. [137] proposed a hybrid modeling approach for forecasting DLR in systems integrating both wind and solar power. They employed a complete ensemble empirical mode decomposition with adaptive noise to decompose the time-series into high-, mediumand low-frequency components. Each component was then modeled using a tailored combination of bidirectional GRU, autoregressive, and multivariable regression models, aligned with the specific behavior of each frequency band.

In summary, the structural dependencies in the data—such as temporal continuity or spatial correlation—serve as critical guidelines for neural network architecture design, and the model's performance can vary significantly depending on how effectively these dependencies are captured and incorporated into the structure.

3.1.3. The Quality of the Data

Real-world collected data often contain missing values, outliers, and variables with unclear relevance to the target prediction task. Such data quality issues degrade model performance and contribute to aleatoric uncertainty, which cannot be reduced through improvements to the model itself [138,139]. Therefore, appropriate pre-processing—such as imputing missing values, removing outliers, and selecting or transforming key features—should be carefully designed [140]. While basic pre-processing typically relies on domain knowledge, there is also a growing number of approaches that adaptively design pre-processing strategies based on the data itself. In this context, attention mechanisms have been introduced to better capture and emphasize informative features [141,142], and encoder–decoder architectures have been employed for latent feature extraction and anomaly [143,144].

Yan et al. [145] addressed the problem of household electricity consumption forecasting by augmenting basic LSTM units with a CNN-based pre-processing stage. Using 1D convolution, they extracted the features directly from raw data and employed a temporal CNN to restructure the input into multi-dimensional batches. This preserved the extracted features in a higher-dimensional format, improving the predictive performance of the model. Zhou et al. [146] were the first to introduce an attention mechanism for PV power generation forecasting, enabling the model to adaptively focus on more informative input features. The works of [147,148] utilized autoencoder (AE) to support feature extraction. Park et al. [147] proposed a model architecture that jointly addresses missing value imputation and load forecasting, allowing robust predictions even in the presence of incomplete data. Specifically, various types of AE were used to encode features which were then fed into an LSTM for joint learning. Also, the work of [149] contributed to anomaly detection in PV generation data by combining a convolutional AE with principal component analysis (PCA). Their approach effectively filtered out anomalies with high accuracy, thereby reducing forecasting errors.

In addition, considering the process by which data is collected can further enhance data quality [150,151]. Song et al. [150] focused on addressing errors in weather forecast data collected from meteorological observation stations. By employing Delaunay triangulation, the study selected observation stations in a structured manner, and utilized a transformer encoder to allow the model to selectively attend to relevant weather stations and meteorological variables. Meanwhile, the work in [151] targeted PV power forecasting in environments lacking high-performance wireless metering modems. To address this, they focused on utilizing nearby PV sample sites. By applying K-means clustering and distance-based sampling, they selected representative sample sites, which significantly improved forecasting performance.

These studies highlight the emergence of various pre-processing strategies designed to systematically address the quality degradation factors that forecasting models encounter in real-world environments—such as missing data, outliers, and data collection methods. In other words, the quality of the data plays a critical role in determining appropriate pre-processing strategies.

Leveraging the quantity, structure, and quality of data can enhance model interpretability, thereby supporting improved generalization performance even under unseen distributions. However, this strength also exposes inherent limitations. When domain knowledge is insufficiently considered during training or model design—for example, shifts in data connectivity due to distributional changes or the presence of adversarial perturbations—highly interpretable models may actually become more vulnerable to critical errors. To address such vulnerabilities, complementary safeguards are required. For instance, anomaly detection techniques can be used to identify adversarial inputs and filter out unreliable predictions, while probabilistic forecasting methods, as will be discussed in Section 3.2, can quantify uncertainty and reduce model overconfidence. Alternatively, one may prioritize data-driven performance by increasing model complexity, allowing the model to adapt even when the target cannot be interpreted. However, as previously mentioned, excessive complexity beyond the informational content of the data may lead to overfitting, ultimately weakening robustness even within the training distribution. Thus, striking an appropriate balance between model complexity and the richness of the data, while ensuring appropriate interpretability, is essential.

3.2. Probabilistic Forecasting for Risk-Aware Decision Making

Probabilistic forecasting refers to the approach by which forecasting models quantify uncertainty, and numerous methodologies have been proposed to support this task [152]. Unlike point forecasting methods that provide a single predicted value, probabilistic forecasting captures the uncertainty inherent in predictions and outputs it in forms such as distributions, prediction intervals, quantiles, or scenarios. This enables a more effective assessment of the range of possible future outcomes and helps decision-making frameworks incorporate the reliability of predictions, thereby moving beyond reliance on deterministic point estimates. Figure 2 provides a worked example of how probabilistic load forecasts flow into UC/OPF formulations. In the conventional deterministic pipeline (top), the point forecasts of system load are used as fixed inputs to UC/OPF, resulting in a single optimal schedule. This approach neglects forecast uncertainty and often leads to infeasible or costly recourse actions when actual conditions deviate from the forecast. In contrast, probabilistic forecasting models (bottom) can generate multiple scenarios of demand. These scenarios are then propagated into scenario-based stochastic UC/OPF, where the optimization problem minimizes the expected operating cost across all scenarios. This formulation explicitly accounts for uncertainty but can incur substantial computational burden.

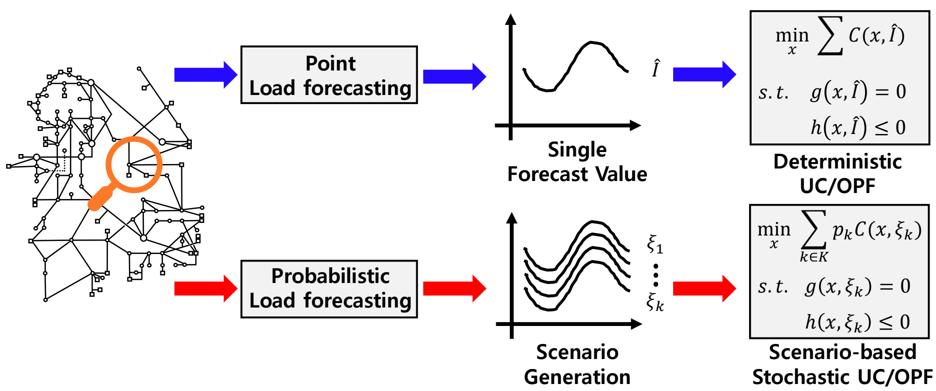

Figure 2. An example of UC/OPF with load probabilistic forecasting.

The evaluation of probabilistic forecasts goes beyond simple accuracy metrics and centers on two key performance criteria: reliability and sharpness [153]. Reliability refers to the consistency between the predicted probabilities and the observed outcomes, while sharpness reflects the concentration of the predictive distribution—narrower intervals are preferred, as long as reliability is maintained. Due to the critical importance of reliability, probabilistic forecasting methods have been developed based on rigorous statistical theory, particularly through two major statistical paradigms: Bayesian and frequentist approaches. The Bayesian approach estimates a posterior distribution based on the prior distribution of the estimator and the likelihood of the observed data. In contrast, the frequentist approach relies on generating repeated samples to construct a sampling distribution [154]. In addition, more scalable and flexible methods—such as quantile regression and deep ensembles—are also widely used, even though they may not provide formal statistical convergence guarantees.

3.2.1. Bayesian Approaches

In deep learning, Bayesian approaches treat model parameters θ not as deterministic values but as random variables, $\theta \sim p(\theta)$, with the goal of estimating the posterior distribution $p(\theta|\mathcal{D})$ conditioned on the observed dataset $\mathcal{D}$. This allows for principled uncertainty quantification by capturing all sources of uncertainty present in both the data and the model. However, the posterior distribution of neural networks is typically high-dimensional and non-convex, making it highly complex. To address this complexity, two main approximation strategies have been developed: variational inference and Monte Carlo dropout [155].

- Variational inference is widely used to approximate the true posterior distribution in Bayesian inference with a tractable parametric distribution [156]. The optimization is performed by minimizing the KL divergence between the true posterior and the chosen parametric distribution, which leads to the derivation of the evidence lower bound (ELBO) as the objective function.However, implementing variational inference typically requires modifications to the model architecture, often resulting in at least a two-fold increase in the number of parameters. In addition, selecting a prior distribution with a practical and well-justified form is critical for achieving meaningful uncertainty estimates.
- Monte Carlo dropout provides an alternative to explicitly estimating the posterior by leveraging dropout layers during inference [157]. By keeping dropout active at test time and performing multiple forward passes, the method approximates Bayesian inference in deep Gaussian processes (GPs). In essence, this approach interprets repeated stochastic forward passes through a dropout-trained model as sampling from the posterior distribution, thereby enabling uncertainty estimation without requiring architectural modifications or explicit posterior modeling.

3.2.2. Frequentist Approaches

Frequentist approaches estimate model uncertainty through empirical distribution of prediction outcomes by a number of samples. This framework includes two main methodologies. The first is ensemble-based, where multiple prediction models are trained on different resampled versions of the training data, and their outputs are aggregated to estimate uncertainty [158]. The second is post hoc, where the predictive distribution is inferred retrospectively by comparing the model's predictions with a held-out, exchangeable dataset that was not used during training [159].

- Resampling-based ensembles are trained using statistical resampling techniques such as jackknife and bootstrap, which are commonly employed to generate multiple training datasets [160,161]. The jackknife method constructs new datasets by systematically leaving out one observation at a time, whereas the bootstrap method operates similarly but allows for sampling with repetition. As a result, the number of resampled datasets increases, enabling smoother and more accurate estimation of predictive distributions.

However, this also raises computational complexity, and such methods have become more practical with the advancement of computing power.
- Conformal prediction is a theoretically grounded and model-agnostic framework that requires no modification to the underlying model, making it broadly applicable across diverse architectures. In the context of neural networks, this approach involves creating a separate calibration dataset that is excluded from model training. Using this calibration set, the model's predictions are compared against the true values to compute nonconformity scores, typically derived from residual (prediction error). The distribution of these scores is then used to construct prediction intervals that offer guaranteed coverage probabilities for unseen data [162]. A key limitation, however, is the requirement for a dedicated calibration set, which can be impractical in data-scarce environments.

3.2.3. Deep-Learning Approaches

Beyond traditional statistical frameworks, several effective methods have been developed for uncertainty quantification. One prominent example is quantile regression, proposed by Koenker and Hallock [163]. This method extends the classical linear regression framework by allowing the estimation of regression curves corresponding to the different percentage points (quantiles) of the target distribution. It restructures the model's output layer to predict multiple quantiles and learns by minimizing the pinball loss, which is specifically designed to handle asymmetric quantile errors. Quantile regression is widely adopted due to its model-agnostic nature and simple implementation. However, because each quantile layer is modeled, respectively, theoretical guarantees for the ordering of quantiles are not enforced, leading to potential issues like quantile crossing.

Another widely used approach is the deep ensemble method [164]. This involves training multiple neural networks with different random initializations and aggregating their predictions to form an empirical predictive distribution. Each network is trained to minimize a loss function based on a proper scoring rule, such as the negative log-likelihood. At inference time, the outputs of the ensemble members are averaged to produce a final predictive distribution, which can be used to quantify uncertainty. When trained with adversarial techniques, the predictive distribution becomes smoother and more robust. As a non-Bayesian method that avoids resampling and retains simplicity, deep ensembles offer a practical and scalable solution for uncertainty quantification.

A variety of approaches have been applied to quantify uncertainty in power system forecasting tasks. Ciaramella et al. [165] and Sun et al. [166] employed Bayesian neural network (BNN) for uncertainty quantification in PV generation and residential net load forecasting, respectively. In both cases, priors were set over network parameters, initialized by incorporating weights and biases from conventional neural networks. AlHakeem et al. [167] combined advanced wavelet decomposition and a generalized regression neural network, optimized via particle swarm optimization, for PV power forecasting. To ensure robustness across seasonal variations, they incorporated bootstrap confidence intervals (CIs) to quantify uncertainty. Similarly, Fan and Hyndman [168] used a modified bootstrap method to estimate load prediction intervals, addressing operational risks and complex seasonality patterns. Zaffran et al. [169] critiqued the exchangeability assumption in standard conformal prediction, arguing that it is unsuitable for time-series data. They proposed adaptive conformal inference, which operates as an online procedure to accommodate temporal dependencies, and demonstrated its use in day-ahead electricity price forecasting by generating valid prediction intervals.

A set of works [170–176] adopted quantile regression due to its model-agnostic nature and ease of implementation. These studies proposed various uncertainty quantification models in the contexts of load and PV forecasting. Brusaferri et al. [177] extended the deep ensemble technique—despite its lack of a formal theoretical foundation—for electricity price forecasting. To strengthen the theoretical foundation of the approach, they incorporated conformal inference and implemented an online recalibration procedure, enabling the quantification of predictive uncertainty for electricity prices. This highlights the practical flexibility of conformal prediction as a post-processing method that can be readily integrated into existing forecasting models.

3.3. Decision-Focused Learning: Incorporating Decision Quality

In applications such as power system operation—where decision-making under uncertainty is critical—it is common to solve an optimization problem based on forecasted parameters. In this context, the output of a forecasting model is not merely a final prediction, but rather serves as a parameter for downstream optimization tasks, directly influencing real-world operational decisions. However, traditional forecasting models are typically trained to minimize prediction error without considering the quality of the resulting decisions or the structure of the optimization problem itself. As a result, it becomes difficult to quantitatively control the impact of forecast errors on downstream decisions, which can lead to degraded overall system performance.

To address this limitation, the decision-focused learning (DFL) paradigm has been proposed. DFL integrates the forecasting model with the downstream optimization problem by directly incorporating the final decision-making process into the training phase, thereby enabling end-to-end learning that jointly considers prediction and optimization. During training, model parameters are updated by minimizing a loss function that reflects the quality of decisions derived from the predictions. This approach enables the model to learn forecasts that are optimized for their ultimate role in supporting high-quality decision making. Figure 3 illustrates how DFL modifies the training pipeline. In conventional learning, the model is trained to minimize only the prediction loss (top), which measures the discrepancy between the prediction and the ground truth. In contrast, DFL is trained to also minimize the regret loss (bottom) by directly incorporating the downstream optimization problem into the training loop. The detailed process is as follows.

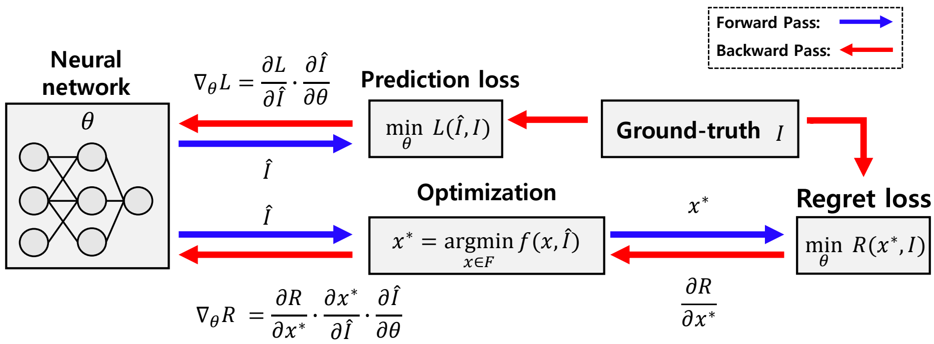

Figure 3. The training process of DFL.

In DFL, the concept of regret is commonly used as a loss function. Let the true parameter to be predicted be denoted as I, the corresponding optimal solution as $x^\star(I)$, and the model's prediction with parameter θ as $\hat{I}(\theta)$. Then, the regret is defined as

$$\text{Regret}\big(x^\star(\hat{I}(\theta)),\, I\big) = \text{Cost}\big(x^\star(\hat{I}(\theta)),\, I\big) - \text{Cost}\big(x^\star(I),\, I\big).$$

This represents the difference between the actual cost of the decision based on the prediction and the ideal cost under full information, serving as a loss function that directly reflects decision quality. In addition to regret, other loss functions such as the cost itself $\mathrm{Cost}\big(x^{\star}(\hat{I}(\theta)), I\big)$ or the decision error $\mathrm{MSE}\big(x^{\star}(\hat{I}(\theta)),\, x^{\star}\big)$ can also be employed.

To perform end-to-end learning based on such loss functions, the following chain rule–based gradient is required:

$$\frac{\partial\,\mathrm{Cost}}{\partial x^{\star}} \cdot \frac{\partial x^{\star}}{\partial \hat{I}} \cdot \frac{\partial \hat{I}}{\partial \theta}.$$

The key challenge lies in computing $\frac{\partial x^{\star}}{\partial \hat{I}}$, i.e., the sensitivity of the optimal solution with respect to the predicted parameters. Standard optimization algorithms do not guarantee the differentiability of the solution, and this issue becomes even more pronounced in MIP problems where the solution space is discrete. Representative approaches to address this challenge include the following [178].

- The analytical differentiation of optimization mappings refers to the approaches that analytically compute the sensitivity of the solution with respect to the optimization problem's parameters. Amos and Kolter [179] proposed OptNet, which introduced the concept of differentiable optimization layers. They demonstrated that the solution to a quadratic program is differentiable with respect to the problem instance by leveraging the Karush–Kuhn–Tucker (KKT) conditions. This idea was further extended by Agrawal et al. [180] through the development of CvxpyLayers, which enables similar differentiability for conic problems. In contrast, Domke [181] and Kotary et al. [182] proposed an alternative approach known as solver unrolling, which embeds the entire optimization algorithm into the computational graph and applies automatic differentiation through it. This method avoids the need for the explicit analytical differentiation of the solution but can be less efficient due to the backward pass being performed over the full algorithmic computation graph.
- Analytical smoothing of optimization mappings involves approximating non-smooth optimization problems to enable gradient-based learning. While linear programming (LP) problems are convex and involve continuous variables, their optimal solutions often lie at the vertices of the feasible region, making them non-differentiable with respect to the input parameters. Wilder et al. [183] addressed this issue by introducing a Euclidean norm regularization term into the objective, effectively smoothing the problem and enabling differentiation. They also demonstrated the effectiveness of replacing integer constraints with LP relaxations when applying DFL to MIP problem. Building on this idea, Ferber et al. [184] improved the relaxation quality by incorporating cutting-plane methods into the approximation of MIP problems. This approach enabled tighter LP relaxations, thereby enhancing the gradient signal for end-to-end learning.
- Smoothing by random perturbations approximates non-differentiable optimization mappings by introducing stochastic perturbations to the input, thereby enabling gradient estimation through differentiable expectations. Domke [185] proposed a finite difference method based on conditional expectations, while Vlastelica et al. [186] approximated gradients using piece-wise linear interpolation. Building on perturbation-based optimization, Berthet et al. [187] introduced differentiable perturbed optimizers by reparametrizing perturbed optimizers [188] with added Gumbel noise to approximate conditional probability distributions. Further developments include the use of multiplicative perturbations [189], Gumbel-softmax distributions [190], and sum-of-gamma distributions [191], all of which enable differentiable approximations of otherwise discrete or non-smooth optimization procedures.

DFL has been increasingly adopted in recent studies on uncertainty forecasting in power systems [192]. Carriere and Kariniotakis [193] applied this approach to renewable power trading in the day-ahead electricity market. Their model jointly predicted renewable generation, day-ahead spot prices, and imbalance prices, with the goal of maximizing overall profit. By designing separate forecasting models for each input and jointly optimizing both their accuracy and the resulting trading strategy, they effectively linked forecasting and operational decisions. The works of [194,195] considered demand forecasting in the context of day-ahead economic dispatch (ED). In particular, Ellinas et al. [195] emphasized that demand forecasting exhibits decision-dependent uncertainty, as demand forecasts ultimately influence generation schedules and costs, which in turn feedback into actual demand. They addressed this complex interaction using Bayesian optimization. Sang et al. [196] introduced a surrogate regret loss to approximate the true regret loss, which is often non-differentiable. The target application, energy storage arbitrage, is formulated as a typical mixed-integer linear program, where the feasible region is a polyhedron. As a result, the optimal solution is likely to lie on an extreme point, making direct differentiation infeasible. To overcome this, they replaced the true regret with a convex upper bound that serves as a tractable surrogate loss.

These works demonstrate the diverse approaches developed for learning to predict in power system operations. To provide a concise overview, representative case studies are summarized in Table 1, while Table 2 compares forecasting paradigms in terms of their strengths, limitations, and evaluation metrics.

Table 1. Summary of case studies for prediction.

Consideration	Method	Model Target	Architecture	Papers
Generalization	Data augmentation	Load	MLP	[109]
	Cross-validation, grid search	PV power	Machine learning	[111]
	Hyperparameter optimization	Electricity price	LSTM	[112]
	Meta-learning	Renewable power	DNN	[118,119]
Structure	Temporal	-	Recurrent architecture	[126–130]
	Spatio-temporal	Residential load	Pooling-based RNN	[131]
		PV Power across multiple sites	CNN with space–time matrix	[132]
			GNN-LSTM	[133]
	Frequency components	PV power	CNN with decomposition	[134]
		Residential load	Residual LSTM	[135]
		Load	Residual MLP	[136]
		DLR, renewable power	Combination of GRU	[137]
Data quality	Feature extraction	Load	CNN-LSTM	[145]
			AE-LSTM	[147,148]
	Anomaly detection	PV power	Convolutional AE with PCA	[149]
	Select data collection sites		Transformer-GRU	[150]
			Attention-RNN	[151]
Error risk	Bayesian	Net load, CIs	Bayesian LSTM	[166]
	Bootstrap	PV power, CIs	Generalized regression	[167]
	Quantile regression	Quantile scenarios	DNN with quantile layers	[170–176]
	Deep ensemble, conformal	Electricity price, CIs	Ensemble of DNN	[177]
Decision	Tractable surrogate regret loss	PV power, market prices	MLP with extreme learning machine	[193]
		Price	Residual MLP	[196]
	Analytical differentiation	Load	MLP	[194]
	Smoothing by random process	Load response to price	GP	[195]

Table 2. Summary of methods for prediction.

Method	Pros	Cons	Typical Metrics
Point forecasting	Simple, widely used in practice, fast to train	No uncertainty of prediction info; errors propagate to decisions	Prediction accuracy (e.g., RMSE, NRMSE, MAE, MAPE)
Probabilistic forecasting	Captures uncertainty of prediction; supports stochastic/robust decision	Higher computational cost, calibration needed	Prediction accuracy, interval reliability and sharpness (e.g., pinball, Winkler score, CRPS)
Decision-focused learning	Directly optimizes downstream decision quality	Requires differentiability or relaxation for regret loss, higher training time	Objective value (e.g., net revenue [193], ED cost [194,195], arbitrage benefits [196])

4. Learning to Surrogate: Replacing Intractable Constraints

To address the challenges discussed in Section 2.3, machine learning-based approaches have recently been proposed. These methods aim to construct surrogate models that learn the underlying system dynamics or stability directly from data, eliminating the need for explicit physical modeling or parameter identification. Furthermore, these approaches also reduce computational complexity and time, facilitating their seamless integration into optimization problems. By training on high-fidelity simulations or real-world measurements, surrogate models can approximate the input–output behavior of complex dynamic systems and serve as computationally efficient replacements.

Surrogate models can be broadly categorized into two main types based on existing studies. The first type involves regression models that directly predict dynamic states or stability-related values. Regression models provide quantitative estimates of stability metrics, allowing for more detailed and fine-grained control within optimization problems. For example, Wu and Wang [197] proposed a transient stability-constrained (TSC)-UC model using an input convex neural network (ICNN). Traditional transient stability assessment requires solving computationally intensive DAEs, which are challenging to integrate into UC problems. By leveraging ICNN, which guarantee convexity and can be encoded as linear programs, the authors approximate the transient stability index (TSI) efficiently and embed it directly into the UC optimization. This approach significantly reduces computational complexity while ensuring that transient stability constraints are met. Similarly, Tuo and Li [198] proposed a rate of change of frequency (RoCoF)-constrained UC model, leveraging an MLP to ensure RoCoF safety in low-inertia power systems. Traditional frequency stability models rely on simplified equivalent system dynamics, which cannot fully capture the nonlinear and high-dimensional behaviors of actual systems. By using the surrogate model embedded with safety threshold constraints, the method efficiently represented RoCoF dynamics and integrated them directly into the UC problem, maintaining computational tractability.

The second type involves classification models that assess the system stability or safety. Unlike regression models, this method does not require manually setting threshold constraints for assessing system stability or safety. Instead, the classification model learns to directly distinguish stable and unstable operating conditions from data, which can simplify the integration of stability assessment into optimization frameworks. In [199,200], an MLP-based classifier was employed to model the feasible region of the SC-OPF problem, which is computationally intensive due to the need to explicitly evaluate a large number of contingency scenarios. By learning to distinguish secure and insecure operating points from data, this approach enables fast and accurate security assessments without requiring the explicit consideration of every contingency, thereby reducing computational effort. While most existing studies focused on either classification or regression approaches, a few recent works have explored hybrid methods that leverage the strengths of both. The study in [201] demonstrated that the complementary use of classification and regression enabled robust stability decisions without the need for manually defined thresholds, while also providing quantitative TSI estimates essential for detailed analysis and optimization. This balanced approach enhanced both the reliability and computational efficiency of transient stability assessment.

Although surrogate models facilitate tractability, they introduce new challenges. As the power system scales up, the surrogate model—particularly neural networks—becomes more complex, potentially increasing the overall optimization burden. Furthermore, their performance heavily relies on prediction accuracy, and their black-box nature limits interpretability, making it difficult for operators to fully trust or understand the results. For instance, system operators may struggle to interpret why a classification model considers an operating point stable, as such models often lack transparency. Representative studies and their methodological characteristics are summarized in Table 3, including frequency-constrained (FC), voltage stability-constrained (VSC), small-signal stability-constrained (SSC), TSC, and SC formulation.

Table 3. Summary of control methods based on embedded surrogate models for power system dynamics and stability.

Papers	Model	Target Dynamics	Model Type	Integration Method	Application
[197]	ICNN	Transient stability	Regressor	Surrogate model embedded with safety threshold constraint	TSC-UC
[198]	MLP	RoCoF	Regressor	Surrogate model embedded with safety threshold constraint	FC-UC
[199]	MLP	N-1 contingency	Classifier	Surrogate model embedded for N-1 contingency security assessment	SC-OPF
[200]	MLP	N-1 contingency and small-signal stability	Classifier	Surrogate model embedded for safety assessment	SC-OPF
[201]	Deep belief network	Transient stability	Classifier and regressor	Surrogate model embedded as constraints	TSC-OPF
[202]	MLP	Voltage stability margin	Regressor	Surrogate model embedded with safety threshold constraint	VSC-OPF
[203]	BNN	Transient stability	Classifier	Surrogate model embedded for safety assessment	SC-OPF
[204]	Multivariate adaptive regression splines	Damping index	Regressor	Surrogate model embedded with safety threshold constraint	SSC-OPF
[205]	Support vector machine (SVM)	Frequency nadir	Classifier	Surrogate model embedded for safety assessment	FC-UC
[206]	MLP	Frequency nadir	Regressor	Surrogate model embedded with safety threshold constraint	FC-UC
[207]	MLP	Frequency nadir	Regressor	Surrogate model embedded with safety threshold constraint	Microgrid scheduling
[208]	SVM	Frequency nadir	Classifier	Surrogate model embedded for safety assessment	FC-UC
[209]	MLP	Frequency	Regressor	MLP-based koopman operator embedded as constraints	Model predictive control (MPC)-based load frequency control
[210]	LSTM and conditional variational AE (CVAE)	Transient dynamics states	Regressor	LSTM and CVAE-based koopman operator embedded as constraints	MPC-based post-contingency stabilization
[211]	MLP	Permanent magnet synchronous machine and flywheel energy storage system (FESS)	Regressor	MLP-based koopman operator embedded as constraints	MPC-based FESS control

5. Learning to Optimize: Neural Solvers for Operational Decision Problems

As explained in Section 2, in large-scale power system operations, decision-making processes—such as UC and OPF—are often formulated as complex optimization problem. When incorporating the elements of uncertainty and system dynamics, these optimization problems become nearly impossible to solve in a single stage from a numerical analysis perspective. Recently, deep learning has emerged as a promising alternative to traditional optimization solvers, aiming to directly approximate the mapping from system states to optimal decisions. This approach, referred to as learning to optimize , offers the potential for significant computational acceleration, making it attractive for real-time applications [212].

Unlike standard machine learning tasks, which primarily focus on optimizing a specific loss function, general physical system operation optimization problems are tightly bound by both optimality and feasibility constraints. Optimality ensures the cost-effectiveness or efficiency of the decisions, while feasibility guarantees that the solutions comply with physical and engineering constraints (e.g., power flow limits, generation bounds). Moreover, modern power systems face topological changes, such as line outages or reconfigurations, demanding that learning-based methods generalize and scale to a variety of system configurations. Therefore, scalability under varying grid topologies is also a crucial consideration. Accordingly, this section outlines how neural networks can be designed to account not only for optimality but also for feasibility and scalability.

5.1. Designing Model Architecture for Scalability

Pan et al. [213] proposed a DNN-based approach where a neural network maps system load inputs to generator outputs, effectively learning the input–decision variable relationship in DC OPF (DCOPF) task. This demonstrates the potential of neural networks in learning to optimize, where models approximate the solution mapping of complex optimization problems. However, as real-world systems become increasingly uncertain and large-scale—due to factors such as distributed energy resources (DERs) and rapidly growing demand from infrastructures like data centers—scalability and reliability have become critical requirements. In response, numerous studies [214–221] have explored neural network-based methods that emphasize self-supervision, optimality, feasibility, and scalability. Each of these perspectives offers distinct advantages, but also comes with trade-offs. We first focus on research motivated by the need for scalable neural architectures in learning to optimize. Considerable effort has been devoted to designing architecture designs that can overcome these trade-offs and support more efficient and generalizable optimization learning frameworks.

Jia et al. [222] proposed a CNN model to solve ACOPF called ConvOPF-DOP, considering topology flexibility. The motivation for using CNN lies in their ability to incorporate topology labels as part of the input, which leads to higher solution accuracy. ConvOPF-DOP takes the topology labels and load data at each bus as an input gives an output of the voltage magnitude and angle of each bus. The authors demonstrated that ConvOPF-DOP achieves high accuracy in solving real-time ACOPF under contingencies, outperforming standard DNN on the IEEE 30-bus system. More recently, Xu et al. [223] proposed a hybrid model called OPFormer that combined CNN and a transformer. By leveraging the transformer's capability to capture long-range dependencies, OPFormer outperformed ConvOPF-DOP in real-time ACOPF with N-1 topology changes on the IEEE 30 and 118-bus systems.

To support reliable and efficient power system operation with the high penetration of DERs, optimization models should account not only for system constraints at a single time step (i.e., snapshot) but also for temporal dynamics and uncertainties. This necessitates extending the traditional OPF formulation to incorporate both uncertainty and multi-period considerations. In this context, Zafar and Chung [224] considered multi-period ACOPF with the renewable energy sources (RES) generator (i.e., wind farm). To solve this in a fast and accurate way, they used an LSTM-RNN to handle time-series inputs, including load demand and renewable generation, and to predict generator outputs and RES injections. However, the test system is the IEEE 39-bus system. Rajaei et al. [225] proposed a deep learning-based alternating direction method of multipliers (ADMM) called ADMM-ML to solve for the large-scale power systems while considering the uncertainty scenarios. In this method, RNN is used to imitate the update procedure in ADMM. Specifically, RNN predicts the Z-update by using the nodal net load for given scenarios, the output of X-update, and the Lagrangian multipliers as input. In doing this, [225] showed that ADMM-ML can accelerate the ADMM computation for solving multi-period ACOPF in IEEE 118-bus and PEGASE 1354-bus systems.

To date, we have reviewed the various forms of ACOPF and their proper neural network model architecture. Recently, however, the GNN model has been studied to represent various optimization problems in a general way. Thanks to the property of GNN, topology flexibility can be naturally satisfied. This is because graph-based models learn at the node level, making them inherently well-suited to handle varying network configurations. Falconer and Mones [226] provided a comprehensive comparison among DNN, CNN, and GNN architectures for topology-changing ACOPF problems, and showed that GNN is particularly effective in handling topological variations. Building on this property, Liu et al. [227] and Yang et al. [228] developed GNN-based transfer learning frameworks to enable the fast adaptation in ACOPF tasks with contingencies. Furthermore, Piloto et al. [229] proposed a GNN model called CANOS that considers scalability. It shows that CANOS can accelerate the computation in large-scale ACOPF with N-1 contingency up to 65 ms while keeping constraints violation low. Meanwhile, [230] established the physical relationship between the procedure of message passing and the Gauss–Seidel iteration of AC power flow. The authors called this model architecture physics-embedded GNN and showed that this can enhance the feasibility of ACOPF solutions.

5.2. Designing Loss Functions for Optimality and Feasibility

To effectively learn decision making policies in power systems, the design of the loss function becomes critical. Learning-based optimization methods typically define a loss function that balances the trade-off between optimality and feasibility. The loss in this context should reflect the structure of the underlying optimization problem, which includes not only a cost objective but also strict physical and operational constraints. Specifically, it is often constructed either by imitating numerical solvers or by directly incorporating the core objectives of the optimization problem.

Consider the learning task as a general constrained optimization problem:

$$x^\star = \arg\min_x f(x) \quad \text{subject to} \quad g_i(x) \le 0,\ h_j(x) = 0,$$

where $f(\cdot)$ is the objective function, and $g_i(i = 1, 2, ...n)$ and $h_j(j = 1, 2, ...m)$ represent inequality and equality constraints, respectively. Let the predicted solution be denoted as $\hat{x}$. We typically design the loss function with three key components:

- Supervised loss: This term encourages the model to imitate the output of an exact solver by minimizing the difference between the predicted decision and the reference solution:

$$\mathcal{L}_{\text{sup}} = \|\hat{x} - x^\star\|.$$

  Since the optimal solution satisfies both optimality and feasibility, minimizing this loss naturally induces these properties and provides a simple training signal. However, generating datasets with solver can be time-consuming or infeasible in highly dynamic environments.

- Objective term: This term minimizes the original objective of the optimization problem, encouraging the model to find decisions that are cost-effective:

$$\mathcal{L}_{\text{obj}} = f(\hat{x}).$$

It directly guides the model to reduce the original problem's objective, thereby promoting optimality.

- Penalty term: To explicitly account for feasibility, penalty terms are added for violating system constraints, where each term is typically weighted based on the constraint violation:

$$\mathcal{L}_{\text{pen}} = \sum_{i=1}^{n} w_i \cdot \max(0, g_i(\hat{x})) + \sum_{j=1}^{m} w_{n+j} \cdot \|h_j(\hat{x})\|.$$

This term allows the model to account for the original problem's constraints, thereby promoting feasibility.

These components are combined into a total loss:

$$\mathcal{L}_{\text{total}} = \alpha_1 \cdot \mathcal{L}_{\text{sup}} + \alpha_2 \cdot \mathcal{L}_{\text{obj}} + \alpha_3 \cdot \mathcal{L}_{\text{pen}}$$

where α_1, α_2, and α_3 are non-negative weights controlling the trade-off among the terms. When the supervised loss is used, the network learns to reproduce decisions that are both optimal and feasible, leveraging the knowledge embedded in solver-generated data. On the other hand, using only the objective and constraint-based terms (i.e., setting $\alpha_1 = 0$) allows for training without explicit labels. This reduces reliance on pre-computed solutions and enables greater scalability, particularly in applications involving dynamic grid topologies or large problem spaces.

5.2.1. Theoretical Foundation: Karush–Kuhn–Tucker Conditions

In constrained optimization, the Lagrangian function provides a principled framework to incorporate constraints into the objective function by introducing a set of auxiliary variables known as Lagrange multipliers [231–233]. These multipliers serve as weights that balance the influence of each constraint on the objective, effectively transforming a constrained problem into an unconstrained one in a higher-dimensional space. The Lagrangian function is defined as

$$\mathcal{L}(x, \lambda, \nu) = f(x) + \sum_{i=1}^{n} \lambda_i g_i(x) + \sum_{j=1}^{m} \nu_j h_j(x)$$

where $\lambda_i, \nu_j \in \mathbb{R}$ are the Lagrange multipliers corresponding to inequality and equality constraints, respectively. Intuitively, these multipliers quantify how much each constraint influences the optimal solution; large multipliers indicate that violating the corresponding constraint would significantly increase the objective value.

At an optimal solution (x^*, λ^*, ν^*), the KKT conditions must hold, providing necessary conditions for optimality [234]:

$$\begin{aligned}
&\nabla_x \mathcal{L}(x^*, \lambda^*, \nu^*) = 0, \quad \text{(stationarity)} \\
&g_i(x^*) \leq 0, \quad \lambda_i^* \geq 0, \quad \lambda_i^* g_i(x^*) = 0, \quad \text{(complementary slackness)} \\
&h_j(x^*) = 0, \quad \text{(primal feasibility).}
\end{aligned}$$

These conditions imply that the optimal solution must simultaneously satisfy feasibility constraints while balancing the objective and constraints via Lagrange multipliers.

This theoretical foundation offers strong motivation. Minimizing a carefully designed loss function—constructed with the specific combination of the objective loss and constraint violation loss described above—can effectively guide the model toward the optimal solution.

5.2.2. Theoretical Foundation: Primal-Dual Method

The KKT conditions provide necessary conditions for optimality, but obtaining or approximating the actual optimal solution requires more specific algorithmic approaches. The primal-dual method is a representative optimization technique for addressing such problems, offering a strategy that simultaneously updates the primal variable x and the dual variables (λ, ν) [235,236]. Specifically, the primal-dual method iteratively performs the following operations until

- The primal variable is updated by minimizing the Lagrangian function with respect to primal variable, while keeping the dual variables fixed.
- Dual variables are updated by maximizing the Lagrangian function, while keeping primal variable fixed. Since the multipliers for inequality constraints must remain non-negative, updates in the negative direction are prevented through projection or similar techniques.

This process is repeated until the KKT conditions are satisfied. This theoretical foundation has inspired a neural network-based solution to search for the specific combination of the objective loss and constraint violation loss.

Building on this theoretical foundation, the approach has been extended to a primal-dual learning framework, where the penalty weights—representing the Lagrange multipliers—are treated as learnable parameters that are adaptively updated during training to reflect the severity of constraint violations. In this framework, the primal variable x (e.g., neural network outputs) is updated to minimize the Lagrangian function. The dual variables λ, ν are updated to maximize it, capturing the extent of constraint violations. This iterative optimization dynamically balances objective cost reduction and constraint satisfaction, improving convergence stability and robustness—particularly in problems with complex or tightly coupled constraints, as commonly found in power systems. Moreover, by integrating dual variable updates into the training loop, the model reduces the need for the manual tuning of penalty weights by automatically adjusting the emphasis on each constraint according to its violation level.

Fioretto et al. [216] introduced a framework combining deep learning with Lagrangian dual methods to predict ACOPF solutions, demonstrating enhanced feasibility by dynamically updating dual variables during training. Donti et al. [217] proposed the DC3 method, which incorporates hard constraints into neural networks by combining penalty terms with primal-dual updates to achieve guaranteed constraint satisfaction. More recently, Kim and Kim [218] developed deep learning-based Lagrangian dual embedding, an unsupervised primal-dual approach that embeds equality constraints directly into the model architecture. Their method demonstrated competitive results in solving ACOPF problems while ensuring constraint satisfaction through primal-dual learning framework.

These approaches highlight the crucial role of Lagrangian theory in bridging classical constrained optimization and modern learning-based methods, providing principled ways to encode feasibility in neural network training.

5.3. Feasibility in Learning-Based Optimization

In learning-based optimization for power systems, ensuring feasibility—compliance with physical and operational constraints—is critically important. If the model output is infeasible, physically meaningless, or even potentially dangerous, it poses a significant barrier to deploying learning-based approaches in safety-critical, real-time power system operations. Traditional optimization solvers explicitly enforce constraints through iterative procedures grounded in rigorous mathematical programming frameworks. In contrast, neural network models, no matter how well the loss function is designed, do not strictly drive the penalty terms to zero. To overcome this challenge, a variety of strategies have been proposed to incorporate feasibility considerations within learning pipelines. These strategies are typically divided into two main categories, depending on when feasibility is enforced during the learning process: post-processing methods vs. in-training methods.

Post-processing methods refine or correct the raw outputs of neural networks to ensure constraint satisfaction. Although they do not guarantee feasibility during the training phase, these approaches offer computationally efficient ways to bridge the gap between learned predictions and the strict operational constraints required in power system optimization. Post-processing methods can be further categorized into three groups: warm-starting conventional solvers, decision subspace prediction, and correction to the feasible set, all of which we will visit shortly.

In contrast, in-training methods aim to ensure or promote feasibility directly during the training process. While post-processing methods provide practical solutions, they leave the underlying learning dynamics untouched, which can result in sub-optimal or unstable predictions. To address these limitations, recent research has focused on embedding constraint satisfaction mechanisms within the training itself. These in-training approaches can be broadly categorized into three methodological directions: worst-case constraint handling, projection layers, and implicit layers—each offering a distinct pathway to integrate feasibility into the learning process.

5.3.1. Warm-Starting Conventional Solvers

Classical optimization solvers are highly sensitive to the choice of initial points, which can significantly affect convergence speed or even lead to non-convergence [237]. One common approach to mitigate this issue leverages the predictive power of neural networks to generate high-quality initial points for classical optimization solvers. By initializing iterative solvers with network outputs close to the optimal solution, the overall solving time can be significantly reduced while preserving the solver's intrinsic ability to enforce feasibility strictly. This hybrid approach effectively combines the speed of learning-based methods with the reliability of mathematical programming techniques.

Diehl [238] proposed a GNN-based approach to warm-start ACOPF solvers, achieving up to 3.75× faster convergence while maintaining solution quality. Cao et al. [239] proposed a fast and explainable warm-start point learning method for ACOPF, using a multi-target binary decision tree with post-pruning to generate interpretable and feasible initializations that accelerate solver convergence. Park et al. [221] proposed a compact learning method that compresses the solution space for ACOPF using principal component analysis. This compression reduces the number of trainable parameters, improving scalability and effectiveness. Their approach enables the warm-starting of exact AC solvers to restore feasibility while significantly speeding up computation.

5.3.2. Decision Subspace Prediction

Instead of predicting the full decision vector, some approaches restrict the neural network output to a subset of variables, while the remaining variables are computed analytically or via deterministic rules that guarantee feasibility. For example, the network may predict only control or discrete variables (such as generator commitments), with continuous states (such as voltages or power flows) subsequently calculated using power flow equations or other physical constraints. This structural decomposition leverages domain knowledge to inherently enforce critical constraints without additional corrections.

Pan et al. [213] proposed a predict-and-reconstruct approach where a neural network predicts generation outputs from load inputs, followed by a reconstruction of phase angles using power flow equations. Huang et al. [214], Jia et al. [222] proposed a method that predicts bus voltages and reconstructs the remaining variables to efficiently solve ACOPF problems. The works [87,240] focused on learning only the binary UC decisions, with the remaining continuous variables solved separately via conventional optimization. Notably, Qin and Yu [87] selected only the subset of binary decisions that are predicted with high confidence or accuracy. Zhu et al. [241] proposed a hybrid approach embedding reinforcement learning within a surrogate Lagrangian relaxation framework, where reinforcement learning agents predict discrete UC decisions and physics-based subproblem solvers handle continuous optimization, achieving near-optimal solutions with significant speedup.

5.3.3. Correction to the Feasible Set

A widely used correction technique is to project the neural network's raw prediction $\hat{x}$ onto the feasible set $\mathcal{F}$ defined by the problem constraints. This is typically formulated as a quadratic optimization problem:

$$x_{\text{proj}} = \arg\min_{x \in \mathcal{F}} \|x - \hat{x}\|_2^2$$

where $\mathcal{F}$ encompasses physical limits such as power flow equations, generation bounds, and voltage constraints. This projection ensures that the final solution strictly satisfies all constraints. However, it may incur a trade-off by introducing sub-optimality relative to the original prediction.

To complement the reconstruction step described earlier, both Pan et al. [213] and Huang et al. [214] applied a post-processing step to enhance feasibility. The work [213] further developed a post-processing procedure based on ℓ_1-projection to ensure that the final solutions satisfy the feasibility constraints of the OPF problem. The work [214] introduced a fast post-processing step specifically designed to enforce box constraints, thereby improving both solution feasibility and computational stability. Liang et al. [242] introduced a homeomorphic projection technique using invertible neural networks to enforce feasibility in constrained optimization, achieving high accuracy and efficiency in non-convex OPF tasks.

5.3.4. Worst-Case Constraint Handling

Some methods explicitly consider worst-case constraint violations during training, resulting in solutions that are more robust to perturbations or modeling errors. Rather than minimizing the expected loss, these approaches optimize for the worst-case scenario over a specified uncertainty set, ensuring that solutions remain feasible under perturbations or modeling errors. While this can make the model overly conservative, it provides strong guarantees on safety and reliability.

Nellikkath and Chatzivasileiadis [243] developed a robust training framework using adversarial sampling. This approach enables the model to produce stable OPF solutions even under worst-case load flow prediction. This led to a reduction in constraint violations, even in the presence of large prediction errors. Venzke et al. [244] developed a framework that provides provable worst-case guarantees for neural network predictions in OPF problems, quantifying maximum constraint violations and sub-optimality while reducing worst-case errors through training on extended input domains. Nellikkath and Chatzivasileiadis [245] proposed a physics-informed neural network approach that integrates AC power flow equations directly into training and explicitly optimizes to minimize worst-case constraint violations, providing rigorous guarantees on output feasibility while maintaining prediction optimality. Zhao et al. [246] proposed a safety-aware training framework that guarantees DNN solution feasibility for optimization problems with linear constraints by calibrating constraints during training, eliminating the need for post-processing.

5.3.5. Implicit Layers

Implicit layers differ from standard feedforward layers in that they define outputs as the solution to an internal optimization or equilibrium condition, rather than a fixed sequence of operations. Unlike approaches that enforce hard constraints via post-processing—such as predicting a subspace of the solution and computing the remaining variables afterward—implicit layers embed this constraint satisfaction process directly into the forward pass. By integrating the constraint-solving mechanism into the architecture itself, feasibility is enforced during training, not just at inference. A key distinction lies in the treatment of backpropagation; instead of computing gradients by traversing each intermediate operation, gradients are obtained analytically via the implicit function theorem. This enables efficient and scalable differentiation through potentially complex and iterative optimization procedures. The approach allows for the tighter integration of structural or physical constraints, often improving model robustness and generalization, albeit at the cost of increased training complexity.

Donti et al. [217] formulated constrained prediction as an implicit differentiable layer that completes partial solutions to satisfy equality constraints and applies gradient-based corrections for inequality constraints, enabling constraint-consistent learning and ensuring feasibility in problems with hard constraints. Nellikkath and Chatzivasileiadis [245] introduced physics-informed neural networks that embed AC power flow equations directly into the training process to ensure feasibility, while rigorously determining and reducing worst-case constraint violations across the entire input domain and maintaining prediction optimality. Ding et al. [247] proposed a reduced policy optimization method that implicitly solves voltage equations by partitioning actions into basic and nonbasic sets, using implicit differentiation for policy updates, and incorporating a reduced-gradient-based projection to enforce equality and inequality hard constraints in continuous control tasks, including smart grid operations. Jia et al. [248] developed a data-driven approach by embedding OptNet layers within a three-stage neural network architecture to efficiently approximate ACOPF problems, representing both inequality and equality constraints, and applying pruning techniques to improve feasibility and computational efficiency. Zeng et al. [249] proposed a general implicit layer framework that uses differentiable KKT conditions to solve convex quadratically constrained quadratic programs, enabling the reliable and efficient learning of feasible ACOPF solutions under constraints. Kim and Kim [218] developed a framework that embeds AC power flow equality constraints directly into neural networks using equation embedding, and applies implicit differentiation with primal-dual training to enforce inequality constraints, thereby guaranteeing feasible solutions with fast convergence and superior optimality in large-scale and time-critical ACOPF problems.

5.3.6. Projection Layers

Projection layers incorporate a projection operation that maps unconstrained network outputs onto a predefined feasible set, ensuring that the outputs satisfy hard constraints explicitly during both training and inference. Formally, given an output y from a previous layer, the projection layer computes $\hat{y} = \Pi_{\mathcal{C}}(y)$, where $\Pi_{\mathcal{C}}$ denotes the Euclidean projection onto the constraint set $\mathcal{C}$. When the feasible set is convex, this projection is well-defined and can be efficiently computed. To enable end-to-end training via gradient-based optimization, gradients are either approximated through smoothing techniques or obtained via the implicit differentiation of the projection operator, which is typically non-differentiable at the boundary.

Donti et al. [217] introduced deep constraint completion and correction, a framework employing differentiable projection layers that complete partial solutions to satisfy equality constraints and apply gradient-based corrections for inequality constraints, ensuring feasibility in deep learning for constrained optimization problems. Kim and Kim [250] developed a projection-aware DNN with a custom projection layer that enforces feasibility in DCOPF, ensuring zero constraint violations and high accuracy with reduced training data. Ding et al. [247] proposed a reduced policy optimization method, which integrates reinforcement learning with the generalized reduced gradient method to handle equality and inequality hard constraints via a reduced-form projection and implicit differentiation, achieving fewer constraint violations—including zero violations in inequality constraints—when applied to OPF with battery energy storage. Chen et al. [251] developed an end-to-end learning framework that integrates DNN with differentiable repair layers acting as projection operators to jointly enforce discrete and continuous constraints in large-scale ED, achieving low optimality gaps without relying on labeled data.

These approaches represent a powerful paradigm shift in learning-based optimization, providing mechanisms to ensure that model outputs remain compliant with critical physical and operational constraints in power system applications. To address the feasibility challenge, a wide range of methods have been developed to either correct infeasible outputs after training or embed constraint enforcement directly into the learning process. Post-processing approaches offer practical and computationally efficient solutions, often benefiting from domain-specific knowledge. However, they do not influence the learning dynamics, potentially leading to sub-optimal or unstable behavior. In contrast, in-training methods aim to ensure feasibility throughout the training process, enabling stronger guarantees on safety and generalization. Although these methods often increased modeling and computational complexity, they represent a promising direction for integrating deep learning with principled optimization frameworks.

5.4. Recipe for Real-Time Use

Building on the theoretical foundations and methodologies presented in Section 5, this section outlines a practical recipe for deploying learning-based optimization in real-time power system operations. In time-critical settings, neural solvers must deliver near-optimal solutions within strict latency budgets while maintaining feasibility and robustness under uncertain operating conditions. Achieving these goals requires a combination of architectural design, algorithmic strategies, and corrective mechanisms that together enable reliable real-time performance.

First, latency management is critical: solution inference and any subsequent corrective procedures must be completed within time windows dictated by system control cycles. This necessitates careful profiling of end-to-end computational delays and a judicious balance between model complexity and post-processing overhead to ensure timely responses. Second, warm-starting strategies provide an effective means of accelerating solver convergence. Neural networks can generate high-quality initial guesses for conventional solvers, allowing hybrid methods to combine the rapid inference of learning-based models with the rigorous feasibility guarantees of classical optimization frameworks. Third, exploiting problem structure through topology-aware architectures further enhances scalability and efficiency. Incorporating network connectivity, sparsity patterns, or hierarchical decompositions through graph neural networks or decomposition-based designs reduces redundancy and preserves fidelity to system physics, facilitating fast and accurate predictions.

Despite these advances, neural models may occasionally produce infeasible or unreliable outputs under previously unseen conditions. To mitigate such risks, real-time frameworks must integrate robust fallback mechanisms that automatically revert to conventional solvers when necessary, ensuring operational continuity and safety. Complementing these strategies, feasibility restoration techniques such as lightweight post-processing, projection layers, reconstruction schemes, and constraint repair heuristics serve as safety nets that correct minor deviations without imposing significant computational burdens.

By integrating these methodologies, learning-based optimization can transition from a research prototype to a deployable solution, effectively bridging the gap between methodological advances and safety-critical, time-sensitive grid operations.

These works illustrate the diverse methodologies developed for learning to optimize in power system operations. To provide a structured overview, Table 4 summarizes representative studies, highlighting the model architectures, loss configurations, and methods used to ensure feasibility.

Table 4. Summary of learning to optimize.

Papers	Model	Loss Configuration	Method for Feasibility
[213]	MLP	Supervised loss, objective term, penalty term	Subspace prediction, correction
[214]	MLP	Supervised loss	Subspace prediction, correction
[215]	MLP	Supervised loss, penalty term	Subspace prediction
[216]	MLP	Supervised loss, objective term, penalty term	Implicit layers
[217]	MLP	Supervised loss, objective term, penalty term	Implicit layers, projection layers
[218]	MLP	Objective term, penalty term	Implicit layers
[219]	MLP	Objective term, penalty term	-
[220]	MLP	Supervised loss, objective term, penalty term	Implicit layers
[221]	MLP	Supervised loss	Warm-starting
[222]	CNN	Supervised loss	Subspace prediction
[223]	CNN, transformer	Supervised loss	Subspace prediction
[224]	LSTM	Supervised loss, objective term	-
[225]	RNN	Supervised loss, objective term, penalty term	Warm starting
[226]	MLP, CNN, GNN	Supervised loss	-
[227]	GNN	Supervised loss, objective term, penalty term	-
[228]	GNN	Objective term, penalty term	Implicit layers
[229]	GNN	Supervised loss, objective term, penalty term	Implicit layers
[230]	GNN	Supervised loss, objective term, penalty term	Implicit layers
[238]	GNN	Supervised loss	Warm starting
[239]	Decision tree	Supervised loss, penalty term	Warm starting
[240]	MLP	Objective term, penalty term	Subspace prediction
[87]	GNN	Supervised loss, objective term	Subspace prediction
[241]	MLP	Objective term, penalty term	Subspace prediction
[242]	-	Objective term, penalty term	Correction
[243]	MLP	Supervised loss	Worst-case handling
[244]	MLP	Supervised loss, objective term, penalty term	Worst-case handling
[245]	MLP	Supervised loss, penalty term	Worst-case handling, Implicit layers
[246]	MLP	Supervised loss, penalty term	Worst-case handling
[247]	MLP	Objective term, penalty term	Implicit layers, projection layers
[248]	MLP	Supervised loss	Implicit layers
[249]	MLP	Supervised loss, penalty term	Implicit layers
[250]	MLP	Supervised loss	Projection layers
[251]	MLP	Objective term, penalty term	Projection layers
[252]	BNN	Supervised loss, objective term, penalty term	-
[253]	GNN	Objective term, penalty term	Implicit layers
[254]	GNN	Objective term, penalty term	-

6. Deployment Challenges for Deep Learning in Power System Operations

To date, we have discussed deep learning for power-system decision making across three roles, i.e., learning to predict, learning to surrogate, and learning to optimize. However, operational readiness depends on more than advanced methodologies. Thus, in this section, we examine four deployment challenges for deep learning in real-world power system operations as follows:

6.1. Hardware and Latency Constraints

Deep learning models for grid operations often require substantial computational power, which can strain existing control center hardware and risk violating strict real-time decision deadlines [255]. High model complexity or large neural networks may introduce latency that is unacceptable for grid operations, where solutions are needed in seconds or subseconds [256]. Mitigation strategies include using hardware accelerators (GPUs/TPUs or even FPGAs) [257] and optimizing models (through pruning or quantization) to accelerate inference [258]. Inference in distributed edge devices or local servers is also explored to reduce communication delays; for example, fog and edge computing frameworks have been used to achieve low-latency control responses in power systems [259]. These approaches help ensure that deep learning-based controllers meet the real-time performance requirements of modern energy management systems.

6.2. Interoperability with SCADA and Energy Management Systems

Integrating deep learning tools into legacy SCADA and energy management systems (EMS) presents significant challenges in interoperability. Traditional SCADA/EMS architectures use vendor-specific protocols and rigid data formats, so seamless interaction with external AI modules is difficult without custom interface development [259]. The lack of standardized frameworks for embedding AI into grid control workflows further complicates integration.

6.3. Regulatory and Compliance Considerations

The power industry is heavily regulated, and the implementation of deep learning in real-time operations must adhere to strict reliability and safety standards. Grid operators have virtually no tolerance for unvetted AI deployments. In other words, any automated decision tool must be thoroughly validated to ensure that it does not jeopardize the reliability of the grid [260]. Compliance requirements require that machine learning-driven decisions be auditable and aligned with existing operational criteria, which can be challenging when using opaque deep neural networks. However, the black-box nature of many deep learning models raises concerns about explainability and accountability; a lack of transparency can erode the trust of regulators and system operators, potentially hindering the approval of AI-based control solutions [261]. In response, researchers and policy makers are focusing on explainable AI techniques and rigorous model governance [262].

6.4. Cybersecurity of Deep Learning Pipelines

Deep learning pipelines in power grid operations introduce new cybersecurity risks that must be managed [263]. Unlike traditional control software, AI models rely on large datasets and complex code dependencies, expanding the attack surface for adversaries [264]. Conventional data integrity and supply chain attacks are a serious concern; for example, attackers could poison training data or manipulate a model update to induce wrong grid control actions [260]. There is also growing alarm that AI systems themselves could be turned against critical infrastructure; experts warn that compromised or adversarial inputs to an operational machine learning model could be used to disrupt grid stability or mask intrusions in industrial control systems [260].

7. Summary and Discussion

The rising complexity and uncertainty in modern power systems—driven by electrification, renewable integration, and evolving consumption behavior—have pushed traditional numerical optimization frameworks to their practical limits. In this context, deep learning has emerged as a powerful enabler for scalable, data-driven decision making in power system operations. This paper has offered a structured review of how neural networks can be strategically integrated into the key components of operational pipelines by classifying their roles into three core functionalities: learning to predict, learning to surrogate, and learning to optimize.

This functional classification serves a dual purpose. On the one hand, it provides a systematic foundation for organizing a diverse and rapidly growing body of literature. On the other hand, it highlights that the design objectives and performance considerations of neural networks differ substantially depending on their role in the decision-making process. For example, predictive models should balance statistical accuracy with the downstream impact of forecast errors on operational outcomes. Surrogate models, while promising for embedding complex physical constraints into optimization problems, should be evaluated not only on approximation quality but also on their complexity and feasibility within real-time decision contexts. Neural solvers, in turn, face the stringent challenge of satisfying hard physical constraints—such as power flow equations and generator limits—within safety-critical systems, demanding specialized architectures and training procedures. Table 5 summarizes these contributions, highlighting the key purposes, design considerations, and practical methods associated with each functional role.

Table 5. Summary of consideration for each functional role.

Function	Objective	Consideration	Methods
Predict	Prediction accuracy	Quantity of data	Data augmentation, meta-learning
		Structure of data	Interpretable model architecture
		Quality of data	Data pre-processing
	Downstream decision	Probabilistic prediction	Variational inference, Monte-Carlo dropout
			Resampling-based ensembles, conformal prediction
			Deep ensembles, quantile regression
		Decision-focused learning	Implicit differentiation
			Tractable surrogate loss
Surrogate	Computational cost	Limiting model complexity	Input convex neural network
	Reliability	Ensuring the upper bound of approximation errors	
Optimize	Optimality	Appropriate loss configuration	Objective, penalty, supervised loss
	Feasibility		Primal-dual learning framework
		Implementation in training	Implicit layers, projection layers
			Worst-case constraint handling
		Post-processing	Warm-starting conventional solvers
			Decision subspace prediction
			Correction to the feasible set
	Scalability	Interpretable model	Structured model architecture
		Pre-train, Fine-tuning	Meta-learning, transfer learning

Compared to other domains—such as computer vision or natural language processing—where foundation models and high-performing general-purpose architectures have already been developed, the power system domain remains in an early stage in terms of the practical integration and deployment of deep learning models. To advance towards foundation model–level capabilities in power systems, a modular and hybrid approach is essential: one that carefully understands the purpose and requirements of each neural component and integrates them with domain-specific knowledge and physical modeling principles. As illustrated in Figure 4, this integration can be conceptualized as a transition from step-by-step decision pipelines toward an integrated module-based architecture. In this framework, neural networks provide three distinct forms of support for final decision making: (i) as the instance of an optimization problem by forecasting exogenous uncertainties (e.g., load, renewable generation, DLR); (ii) as part of the problem formulation by approximating complex physical constraints and dynamics; and (iii) as the solver that directly produces operational decisions. By configuring forecasting, surrogate modeling, and optimization around a common decision purpose, these modules can be composed into a scalable and domain-informed architecture for power system operation.

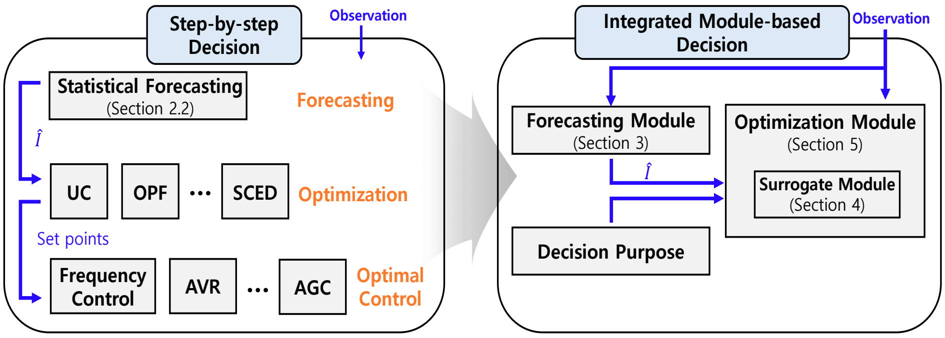

Figure 4. Transition from conventional operation to integrated module framework.

In this regard, the contributions of this review lie not only in synthesizing existing methodologies but also in framing a structured path forward. By clarifying the distinct functional roles neural networks can play, and the technical considerations each role entails, we hope that this work offers practical guidance for designing learning-based frameworks that are both principled and deployable in real-world power system operations. Specifically, Table 6 provides a comparative discussion of the three roles across key aspects of deep learning. This discussion underscores that prediction, surrogate, and optimization models face fundamentally different challenges, and recognizing these distinctions will help advance each role more effectively while also supporting their integration into unified decision-making frameworks.

Table 6. Comparative discussion of neural network roles toward future development.

Aspect	Role	Discussion
Data requirements	Predict	Large volumes of observed, historical measurements (e.g., load/DER, weather, market)
	Surrogate	High-fidelity simulation outputs or precisely measured physical states (e.g., supervisory control and data acquisition)
	Optimize	Numerous problem instances to learn feasible and optimal mappings (e.g., diverse load or renewable perturbation scenarios)
Scalability	Predict (flexible)	Scalable to high-dimensional inputs, as larger data and model capacity can be leveraged without strict structural limits
	Surrogate (restricted)	Scalability limited since the model is embedded within optimization, requiring controlled complexity to remain tractable
	Optimize (moderate)	Needs to scale across diverse system sizes and topologies, but overall runtime should still meet real-time operational limits
Interpretability	Predict (high)	Substantial progress with interpretable forecasting models
	Surrogate (early-stage)	Actively explored through supervised regression/classification models, with growing opportunities for enhancing interpretability
	Optimize (emerging)	While some approaches remain supervised black-box solvers, recent work explores interpretability through structured losses and architecture designs

Table 6. *Cont.*

Aspect	Role	Discussion
Real-world readiness	Predict (deployed)	Widely used in industry forecasting systems with proven operational value
	Surrogate (early-stage)	Promising in research, but practical deployment remains rare due to reliability and tractable
	Optimize (careful)	Actively studied on large-scale test systems, but because decisions directly affect real power system operation, real-world integration must proceed cautiously and remains limited for now
Typical failure	Predict	Distribution shift or overfitting; overall impact is moderate, though forecast errors can propagate into downstream optimization
	Surrogate	Approximation errors may accumulate and render the embedded optimization procedure unreliable or even infeasible
	Optimize	Failure to achieve faster inference than conventional solvers, or degraded optimality compared to solver-based benchmarks
Mitigation strategies	Predict (Robustness)	Anomaly detection, probabilistic forecasting, and decision-focused learning to reduce vulnerability to distribution shifts
	Surrogate (Error bound)	Physics-informed modeling and hybrid approaches to ensure bounded approximation errors and improve reliability
	Optimize (Correction)	Appropriate post-processing such as warm-starting conventional solvers or projection/correction to restore feasibility and optimality
Performance metric	Predict	Deterministic forecast accuracy (e.g., MAE, RMSE, MAPE, NMAE, NRMSE, R-squared score)
		Probabilistic forecast accuracy (e.g., CRPS, pinball loss, Winkler score, calibration plots)
		Downstream objective value (e.g., generation cost, arbitrage profit, unit commitment cost)
		Robustness to worst-case scenarios (e.g., cost under worst prediction)
	Surrogate	Approximation fidelity to physics-based models (e.g., RMSE, MAE, probability of violating the surrogate constraints)
		Computational complexity (e.g., average CPU time, number of variables and constraints)
		Stability-oriented evaluation (e.g., maximum RoCoF, stability under contingency simulations)
	Optimize	Relative objective gap compared with conventional solvers (e.g., generation cost, dispatch cost, unit commitment cost)
		Inequality/equality constraint violations (e.g., percentage of solutions that satisfy the power flow equation)
		Computational time (e.g., inference time compared with commercial solvers, total runtime including post-processing)
		Applicability to larger system size, new topologies or operating conditions (e.g., test-system size, fine-tuning training time)

Building on the functional understanding presented in this review, a promising future direction lies in the development of modular and hybrid architectures that integrate multiple neural roles—prediction, surrogate modeling, and optimization—within a unified framework. Rather than designing isolated models for each task, future research should aim to construct end-to-end pipelines in which each neural component is jointly optimized for overall system reliability and efficiency, with clear interfaces and compatibility among modules. This composability is essential for scaling learning-based decision frameworks to increasingly complex power systems.

As deep learning research for power systems becomes more advanced, there is an opportunity to pursue general-purpose, pre-trained models that can be adapted to a wide range of operational tasks across different system topologies and temporal horizons. By leveraging transfer learning, meta-learning, and multi-task learning techniques, such models could evolve into domain-specific foundation models for power system operations. However, realizing this vision will require the careful balancing of physical fidelity, constraint satisfaction, and computational tractability—challenges that demand continued collaboration between deep learning and power engineering communities. For reproducibility and to facilitate future research, Appendix A provides a collection of publicly available datasets, open-source frameworks, and test systems referenced throughout this review.

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Abbreviations

The following abbreviations are used in this manuscript:

AI	Artificial intelligence
IEA	International energy agency
ENTSO-E	European Network of Transmission System Operators for Electricity
STEPS	Stated policies scenario
SDGs	Sustainable Development Goals
DERs	Distributed energy resources
RES	Renewable energy sources
PV	Photovoltaic
OPF	Optimal power flow
UC	Unit commitment
ED	Economic dispatch
AC	Alternating current
DC	Direct current
ACOPF	AC OPF
DCOPF	DC OPF
SC	Security-constrained
TSC	Transient stability-constrained
FC	Frequency-constrained
VSC	Voltage stability-constrained
SSC	Small-signal stability-constrained
RoCoF	Rate of change of frequency
TSI	Transient stability index
SCADA	Supervisory control and data acquisition
PMU	Phasor measurement unit
DLR	Dynamic line rating
FESS	Flywheel energy storage system
MIP	Mixed-integer programming

LP	Linear programming
DAEs	Differential-algebraic equations
ODE	Ordinary differential Equation
ELBO	Evidence lower bound
KKT	Karush–Kuhn–Tucker
MPC	Model predictive control
DNN	Deep neural network
MLP	Multi-layer perceptron
RNN	Recurrent neural network
GRU	Gated recurrent unit
LSTM	Long short-term memory
CNN	Convolutional neural network
GNN	Graph neural network
BNN	Bayesian neural network
ICNN	Input convex neural network
AE	Autoencoder
CVAE	Conditional variational AE
SVM	Support vector machine
PCA	Principal component analysis
GP	Gaussian process
ADMM	Alternating direction method of multipliers
DFL	Decision-focused learning
CIs	Confidence intervals
MSE	Mean squared error
MAE	Mean absolute error
NMAE	Normalized MAE
RMSE	Root MSE
NRMSE	Normalized RMSE
MAPE	Mean absolute percentage error
CRPS	Continuous ranked probability score

Appendix A. Public Datasets, Open Source Code

This appendix provides a collection of publicly available datasets, benchmark tasks, surrogate modeling frameworks, and standard test systems, organized by category, to facilitate practical use and reproducibility.

Table A1. Datasets and open source codes for prediction.

<table>
<tr><th>Dataset</th><th>Data List</th><th>Horizon</th><th>Model</th><th>Code</th></tr>
<tr><td rowspan="3">HETCom2024 [265]</td><td>Wind</td><td rowspan="3">3 years (2020–2023)</td><td>Quantile regression</td><td>[266]</td></tr>
<tr><td>Solar</td><td rowspan="2">LightGBM</td><td rowspan="2">[267]</td></tr>
<tr><td>Price</td></tr>
<tr><td rowspan="4">GEFCom2014 [268]</td><td>Load</td><td>5 years</td><td>Generalized additive models</td><td rowspan="4">-</td></tr>
<tr><td>Price</td><td>3 years (2011–2013)</td><td>Generalized additive models</td></tr>
<tr><td>Wind</td><td>2 years (2012–2014)</td><td>Gradient boosting machines</td></tr>
<tr><td>Solar</td><td>2 years (2012–2014)</td><td>Gradient boosting, k-nearest neighbor</td></tr>
</table>

Table A1. *Cont.*

Dataset	Data List	Horizon	Model	Code
NREL-PV [269]	Solar	1 year (2006)	STCNN	[270]
			AnyCast	-
NREL-NSRDB [271]	Solar irradiance	18 years (1998–2016)	CGAE	-
-	Solar	1 year (2018–2019)	Natural gradient boosting	[272]
-	Solar	2 years (2019–2021)	Transformer, GRU	[273]

Table A2. Surrogate model frameworks and applications.

Code	Model	Application
[274]	MLP, RNN, GP	Python (3.7, 3.8, 3.9)toolbox enabling surrogate models to be embedded in MPC and estimation problems
[275]	MLP, CNN, neural ODE, PINN, Koopman networks	PyTorch (1.13.1) framework for end-to-end differentiable predictive control with surrogate dynamical models
[276]	GP, radial basis function, random forest, etc.	Benchmarking surrogate-based optimization algorithms on expensive black-box problems
[277]	Radial basis function	Educational MATLAB (R2018b) toolbox illustrating the workflow of surrogate modeling-based optimization
[278]	MLP	Surrogate model for grid-following inverter

Table A3. Public test systems for learning to optimize.

Public Test System	Country	Data Type
[279]	USA	Snapshot
[280]	Poland	Snapshot
[281]	Brazil	Snapshot
[282]	Spain	Sanpshot
[283]	USA	Snapshot
[284]	USA	Spatio-temporal

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